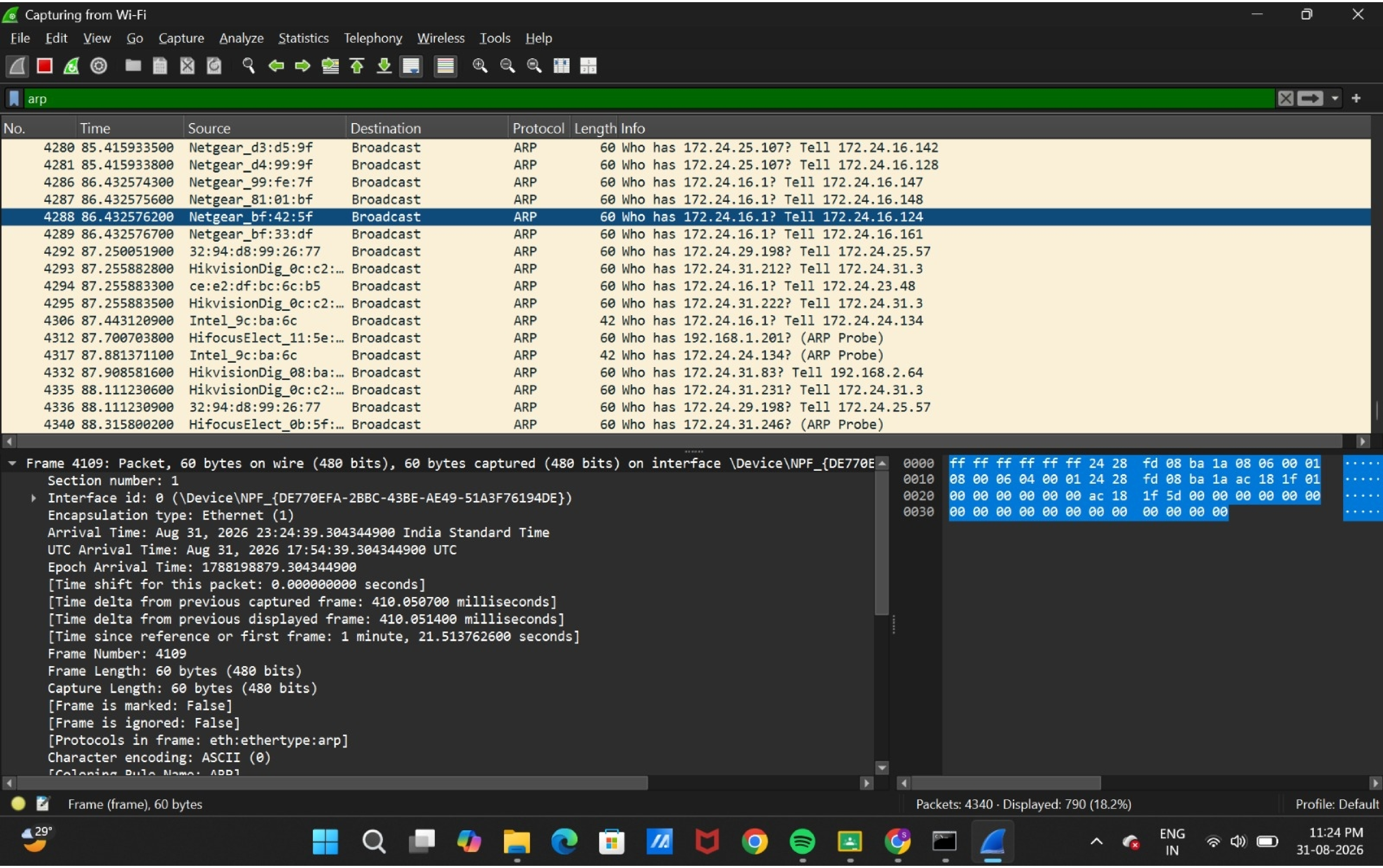
Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <string.h>

int main() {

    char message[100];

    printf("==================================\n");
    printf("   TCP ECHO CLIENT-SERVER\n");
    printf("==================================\n");

    while (1) {

        // Client enters message
        printf("\nClient: ");
        fgets(message, sizeof(message), stdin);

        // Remove newline
        message[strcspn(message, "\n")] = '\0';

        // Client sends message
        printf("Client sends  : %s\n", message);

        // Server receives message
```

input

```
Connection closed.

TCP Echo simulation completed successfully.


...Program finished with exit code 0
Press ENTER to exit console.
```



Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
arp
No. Time Source Destination Protocol Length Info
4280 85.415933500 Netgear_d3:d5:9f Broadcast ARP 60 Who has 172.24.25.107? Tell 172.24.16.142
4281 85.415933800 Netgear_d4:99:9f Broadcast ARP 60 Who has 172.24.25.107? Tell 172.24.16.128
4286 86.432574300 Netgear_99:fe:7f Broadcast ARP 60 Who has 172.24.16.1? Tell 172.24.16.147
4287 86.432575600 Netgear_81:01:bf Broadcast ARP 60 Who has 172.24.16.1? Tell 172.24.16.148
4288 86.432576200 Netgear_bf:42:5f Broadcast ARP 60 Who has 172.24.16.1? Tell 172.24.16.124
4289 86.432576700 Netgear_bf:33:df Broadcast ARP 60 Who has 172.24.16.1? Tell 172.24.16.161
4292 87.250051900 32:94:d8:99:26:77 Broadcast ARP 60 Who has 172.24.29.198? Tell 172.24.25.57
4293 87.255882800 HikvisionDig_0c:c2:… Broadcast ARP 60 Who has 172.24.31.212? Tell 172.24.31.3
4294 87.255883300 ce:e2:df:bc:6c:b5 Broadcast ARP 60 Who has 172.24.16.1? Tell 172.24.23.48
4295 87.255883500 HikvisionDig_0c:c2:… Broadcast ARP 60 Who has 172.24.31.222? Tell 172.24.31.3
4306 87.443120900 Intel_9c:ba:6c Broadcast ARP 42 Who has 172.24.16.1? Tell 172.24.24.134
4312 87.700703800 HifocusElect_11:5e:… Broadcast ARP 60 Who has 192.168.1.201? (ARP Probe)
4317 87.881371100 Intel_9c:ba:6c Broadcast ARP 42 Who has 172.24.24.134? (ARP Probe)
4332 87.908581600 HikvisionDig_08:ba:… Broadcast ARP 60 Who has 172.24.31.83? Tell 192.168.2.64
4335 88.111230600 HikvisionDig_0c:c2:… Broadcast ARP 60 Who has 172.24.31.231? Tell 172.24.31.3
4336 88.111230900 32:94:d8:99:26:77 Broadcast ARP 60 Who has 172.24.29.198? Tell 172.24.25.57
4340 88.315800200 HifocusElect_0b:5f:… Broadcast ARP 60 Who has 172.24.31.246? (ARP Probe)
Frame 4109: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{DE770E
Section number: 1
Interface id: 0 (\Device\NPF_{DE770EFA-2BBC-43BE-AE49-51A3F76194DE})
Encapsulation type: Ethernet (1)
Arrival Time: Aug 31, 2026 23:24:39.304344900 India Standard Time
UTC Arrival Time: Aug 31, 2026 17:54:39.304344900 UTC
Epoch Arrival Time: 1788198879.304344900
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 410.050700 milliseconds]
[Time delta from previous displayed frame: 410.051400 milliseconds]
[Time since reference or first frame: 1 minute, 21.513762600 seconds]
Frame Number: 4109
Frame Length: 60 bytes (480 bits)
Capture Length: 60 bytes (480 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:arp]
Character encoding: ASCII (0)
0000 ff ff ff ff ff ff 24 28 fd 08 ba 1a 08 06 00 01
0010 08 00 06 04 00 01 24 28 fd 08 ba 1a ac 18 1f 01
0020 00 00 00 00 00 00 ac 18 1f 5d 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00
Frame (frame), 60 bytes
Packets: 4340 · Displayed: 790 (18.2%)
Profile: Default
29°
ENG
IN
11:24 PM
31-08-2026

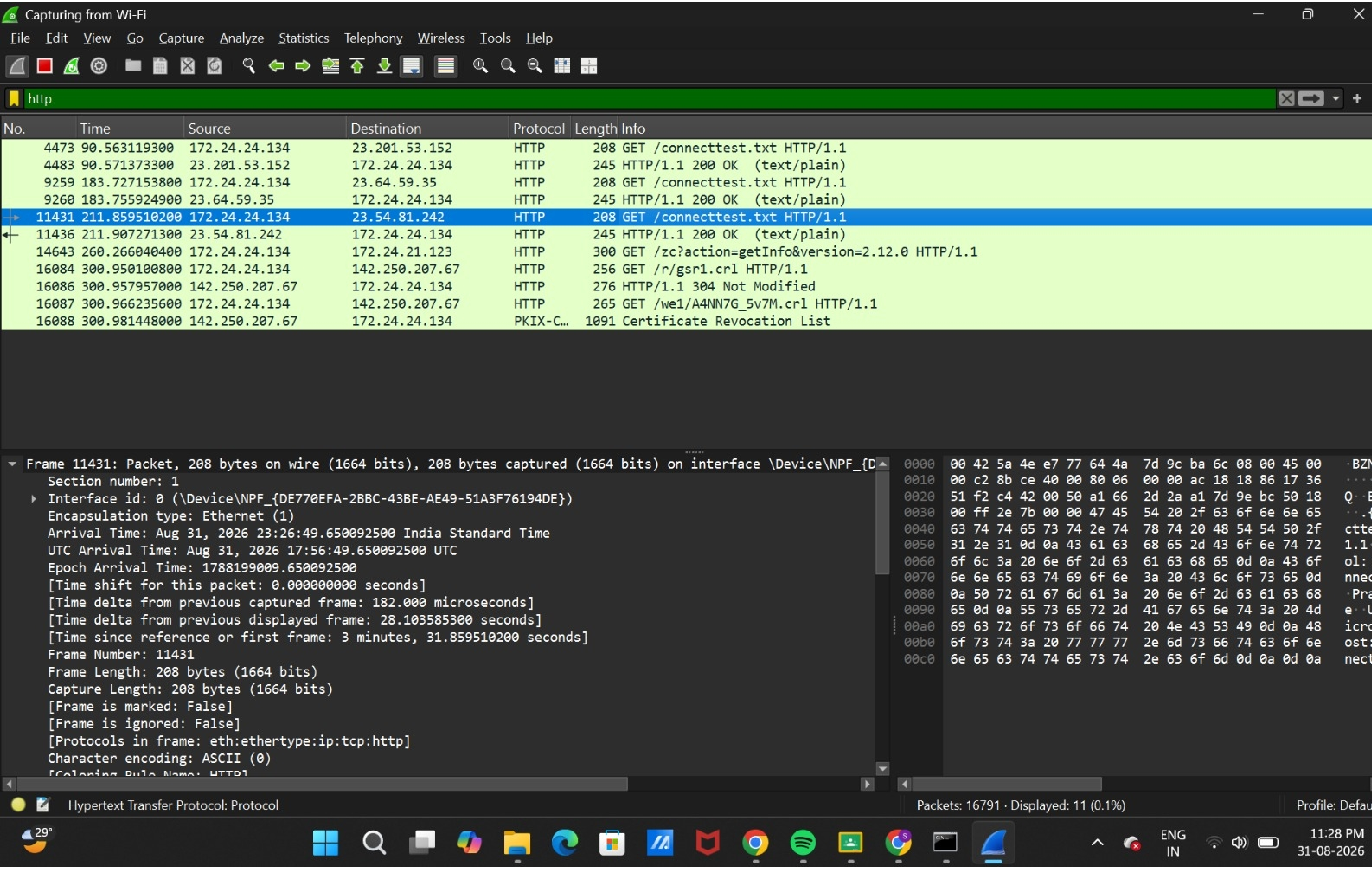
Run | Debug | Stop | Share | Save | { } Beautify

main.c

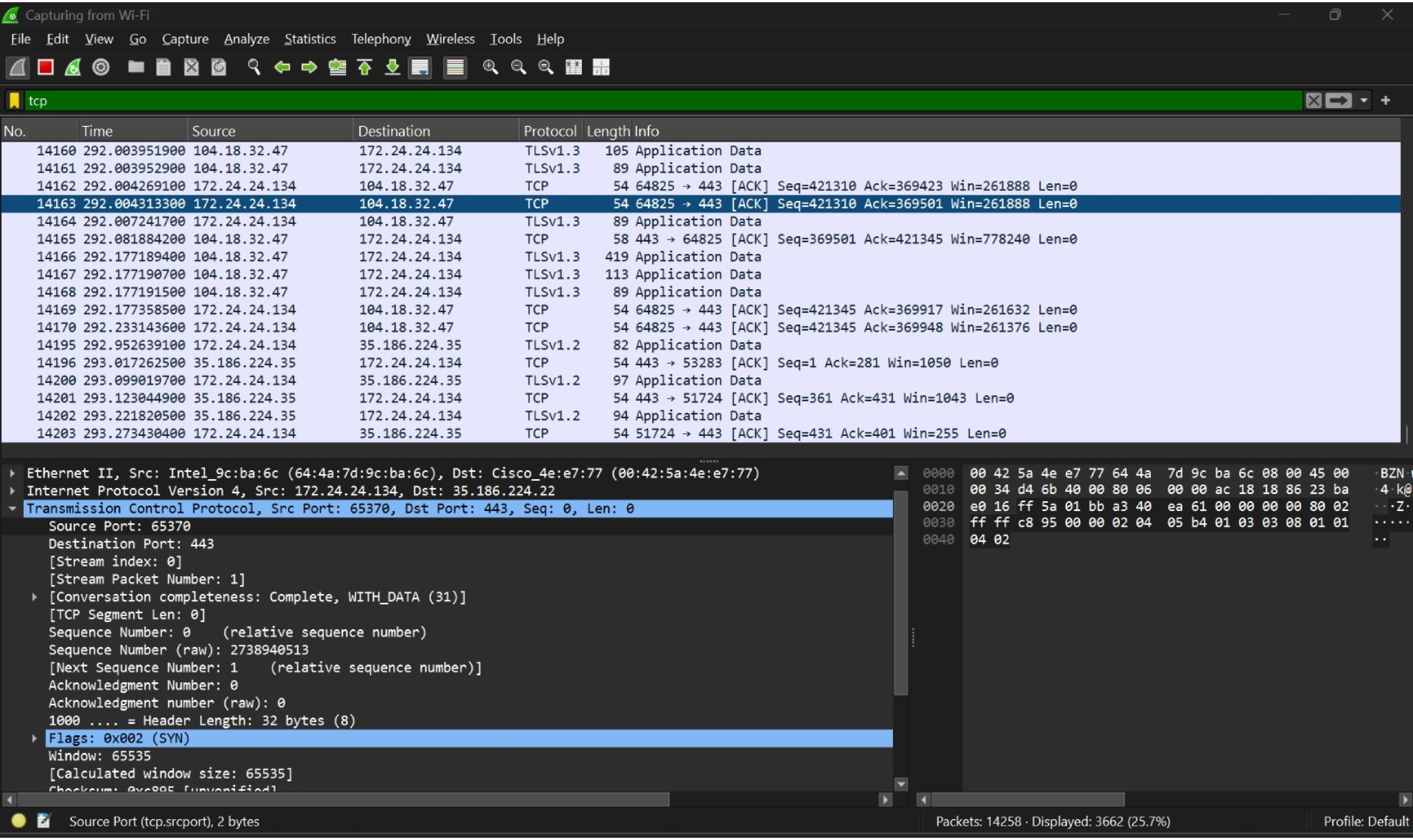
```c
#include <stdio.h>
#include <string.h>

int main() {
    char data[100];
    char stuffed[200];

    int i, j = 0;
    int count = 0;

    printf("Enter binary data: ");
    scanf("%s", data);

    for (i = 0; i < strlen(data); i++) {

        stuffed[j++] = data[i];

        if (data[i] == '1') {
            count++;

            if (count == 5) {
```

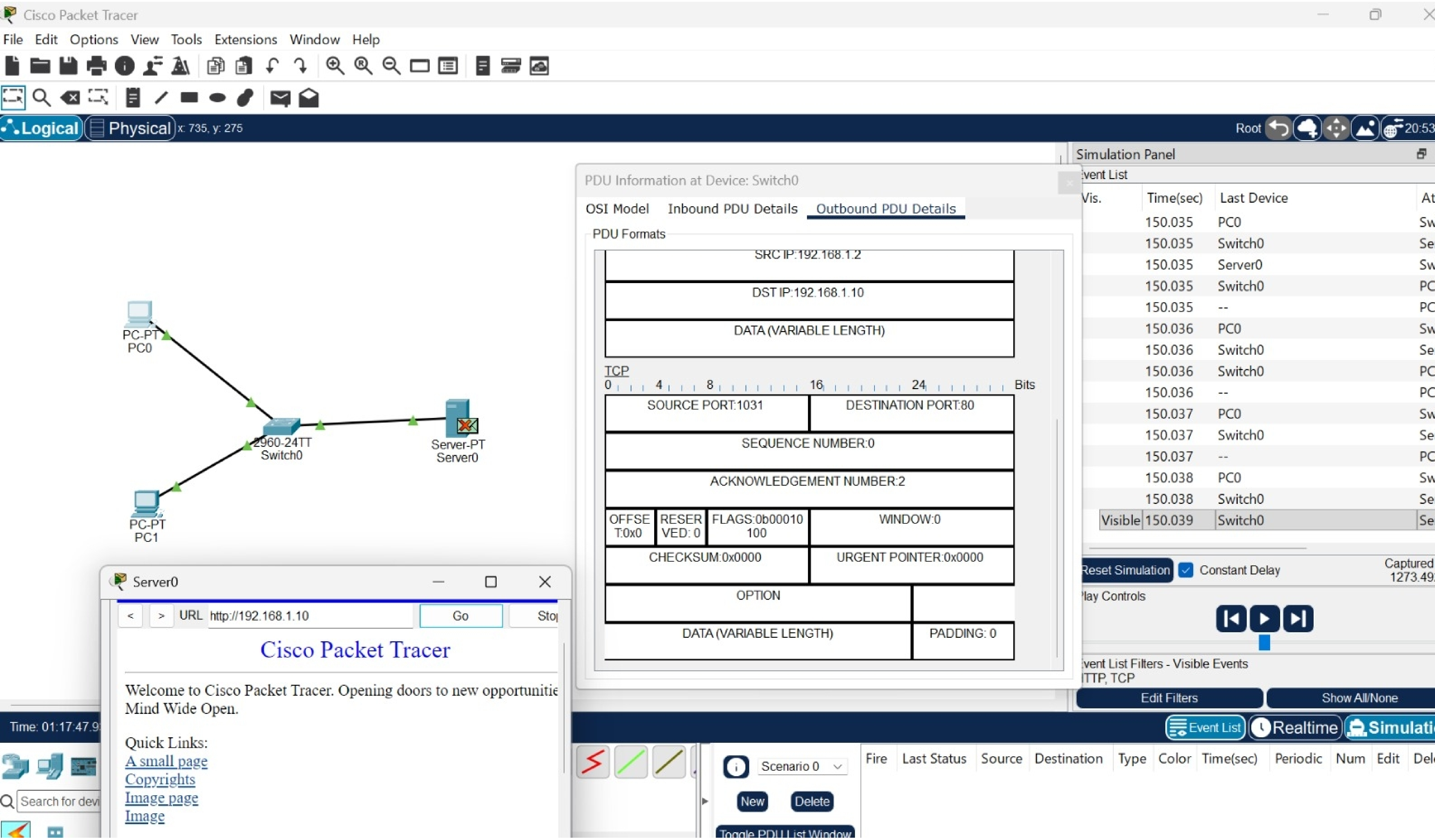
input

```
Enter binary data: 01111110
Original Data : 01111110
Stuffed Data  : 011111010


...Program finished with exit code 0
Press ENTER to exit console.
```

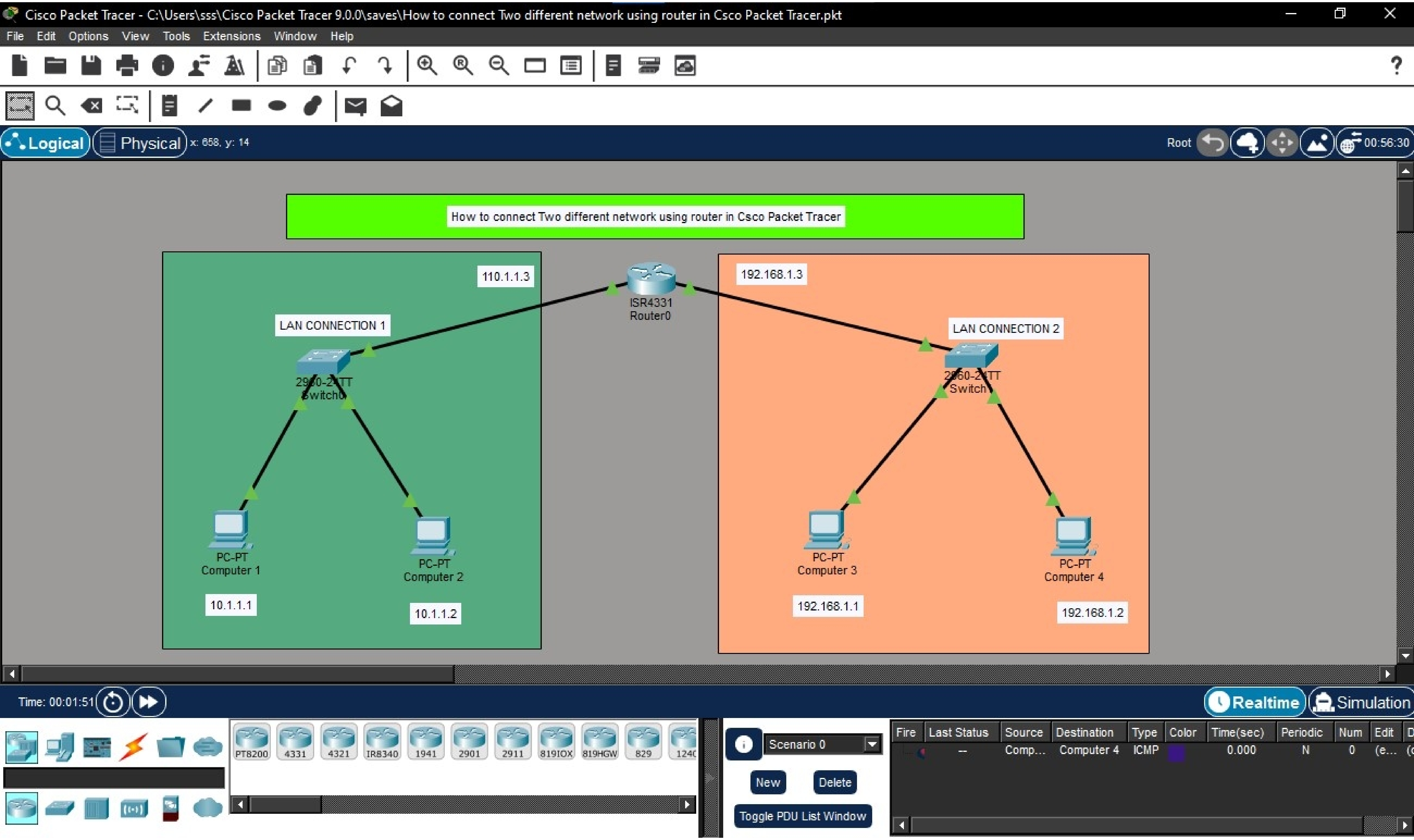


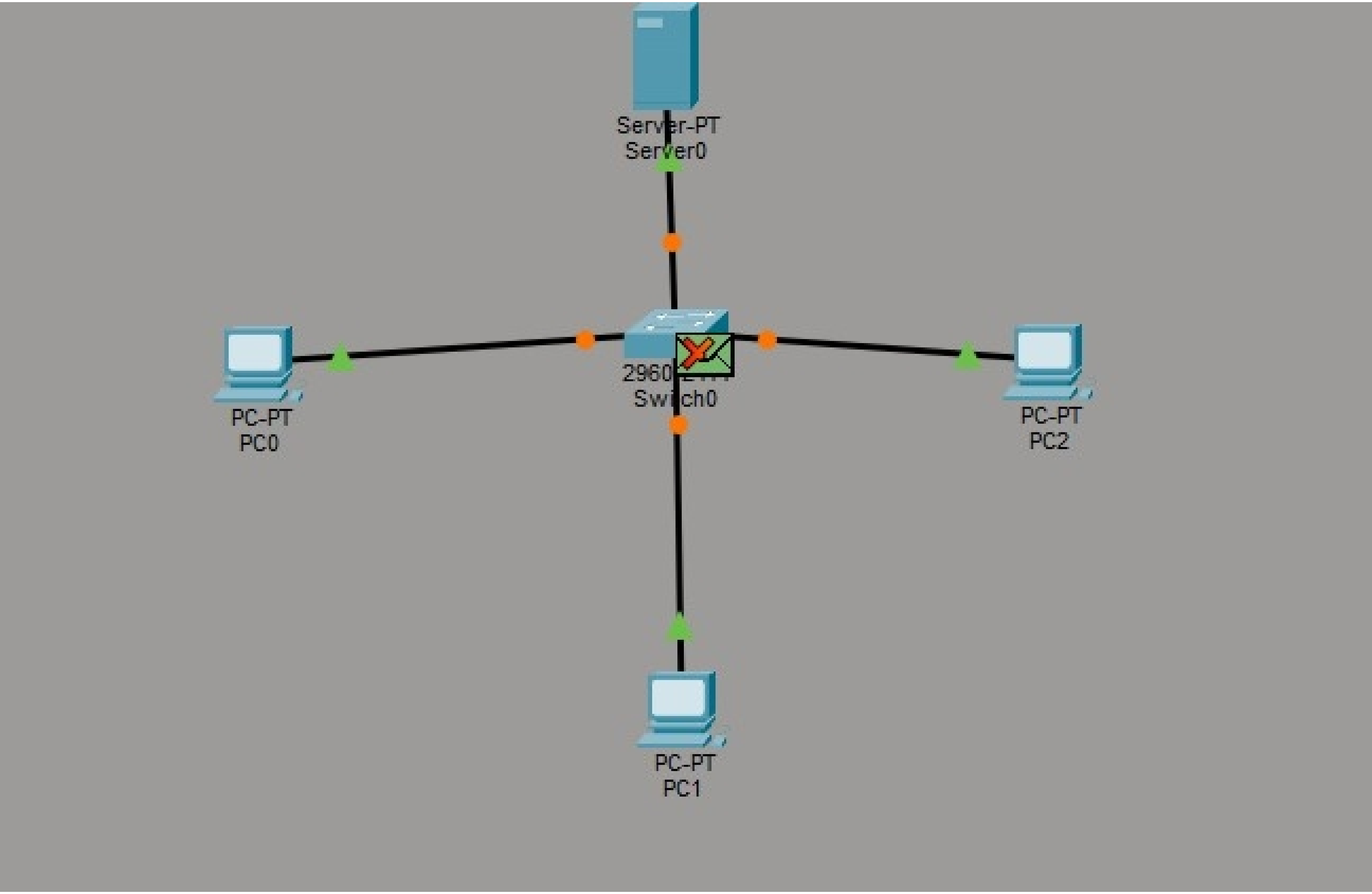
Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
4473 90.563119300 172.24.24.134 23.201.53.152 HTTP 208 GET /connecttest.txt HTTP/1.1
4483 90.571373300 23.201.53.152 172.24.24.134 HTTP 245 HTTP/1.1 200 OK (text/plain)
9259 183.727153800 172.24.24.134 23.64.59.35 HTTP 208 GET /connecttest.txt HTTP/1.1
9260 183.755924900 23.64.59.35 172.24.24.134 HTTP 245 HTTP/1.1 200 OK (text/plain)
11431 211.859510200 172.24.24.134 23.54.81.242 HTTP 208 GET /connecttest.txt HTTP/1.1
11436 211.907271300 23.54.81.242 172.24.24.134 HTTP 245 HTTP/1.1 200 OK (text/plain)
14643 260.266040400 172.24.24.134 172.24.21.123 HTTP 300 GET /zc?action=getInfo&version=2.12.0 HTTP/1.1
16084 300.950100800 172.24.24.134 142.250.207.67 HTTP 256 GET /r/gsr1.crl HTTP/1.1
16086 300.957957000 142.250.207.67 172.24.24.134 HTTP 276 HTTP/1.1 304 Not Modified
16087 300.966235600 172.24.24.134 142.250.207.67 HTTP 265 GET /we1/A4NN7G_5v7M.crl HTTP/1.1
16088 300.981448000 142.250.207.67 172.24.24.134 PKIX-C… 1091 Certificate Revocation List
Frame 11431: Packet, 208 bytes on wire (1664 bits), 208 bytes captured (1664 bits) on interface \Device\NPF_{D
Section number: 1
Interface id: 0 (\Device\NPF_{DE770EFA-2BBC-43BE-AE49-51A3F76194DE})
Encapsulation type: Ethernet (1)
Arrival Time: Aug 31, 2026 23:26:49.650092500 India Standard Time
UTC Arrival Time: Aug 31, 2026 17:56:49.650092500 UTC
Epoch Arrival Time: 1788199009.650092500
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 182.000 microseconds]
[Time delta from previous displayed frame: 28.103585300 seconds]
[Time since reference or first frame: 3 minutes, 31.859510200 seconds]
Frame Number: 11431
Frame Length: 208 bytes (1664 bits)
Capture Length: 208 bytes (1664 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:http]
Character encoding: ASCII (0)
0000 00 42 5a 4e e7 77 64 4a 7d 9c ba 6c 08 00 45 00
0010 00 c2 8b ce 40 00 80 06 00 00 ac 18 18 86 17 36
0020 51 f2 c4 42 00 50 a1 66 2d 2a a1 7d 9e bc 50 18
0030 00 ff 2e 7b 00 00 47 45 54 20 2f 63 6f 6e 6e 65
0040 63 74 74 65 73 74 2e 74 78 74 20 48 54 54 50 2f
0050 31 2e 31 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72
0060 6f 6c 3a 20 6e 6f 2d 63 61 63 68 65 0d 0a 43 6f
0070 6e 6e 65 63 74 69 6f 6e 3a 20 43 6c 6f 73 65 0d
0080 0a 50 72 61 67 6d 61 3a 20 6e 6f 2d 63 61 63 68
0090 65 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d
00a0 69 63 72 6f 73 6f 66 74 20 4e 43 53 49 0d 0a 48
00b0 6f 73 74 3a 20 77 77 77 2e 6d 73 66 74 63 6f 6e
00c0 6e 65 63 74 74 65 73 74 2e 63 6f 6d 0d 0a 0d 0a
Hypertext Transfer Protocol: Protocol
Packets: 16791 · Displayed: 11 (0.1%)
Profile: Default
29°
ENG
IN
11:28 PM
31-08-2026

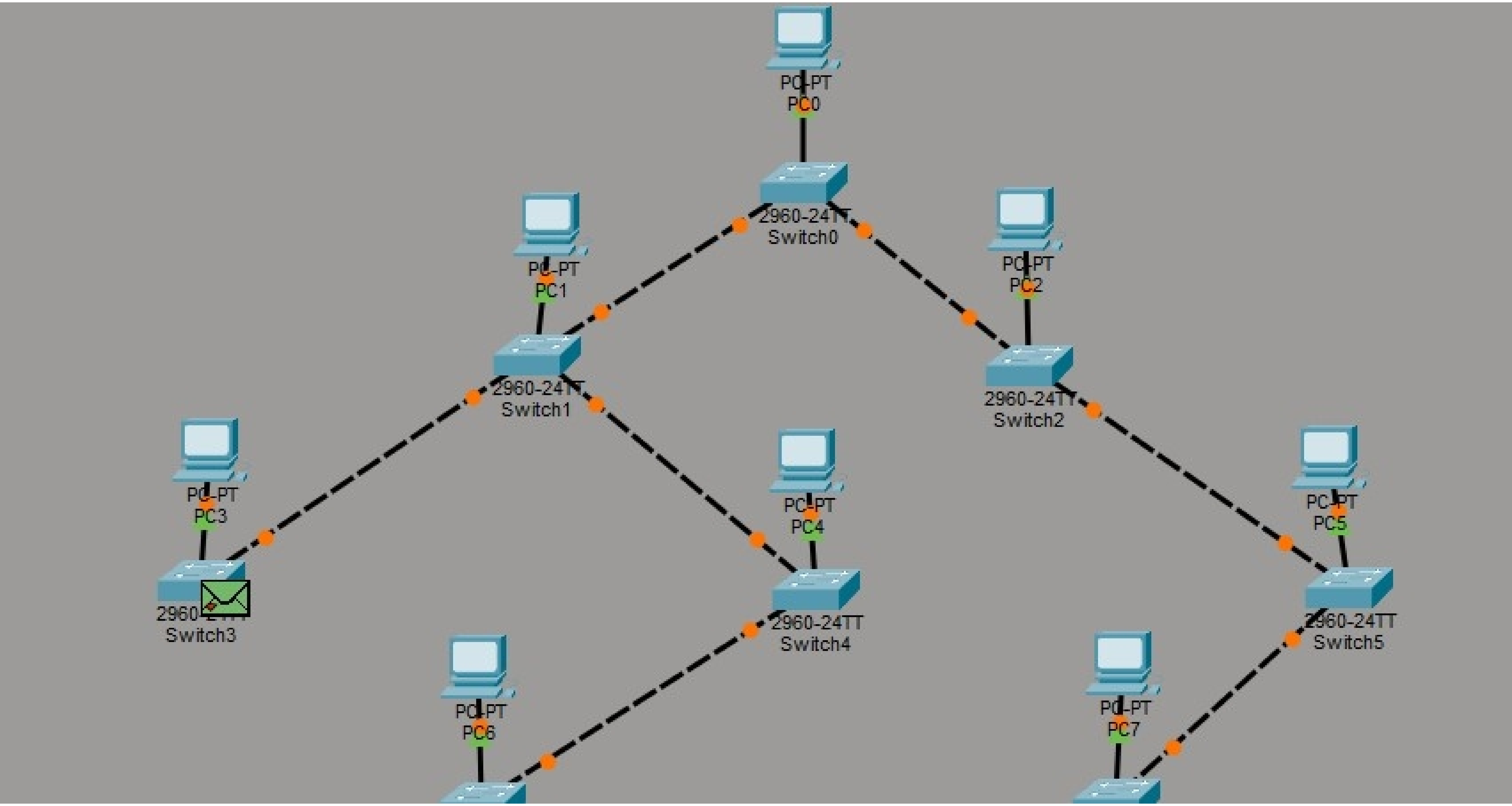

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
14160 292.003951900 104.18.32.47 172.24.24.134 TLSv1.3 105 Application Data
14161 292.003952900 104.18.32.47 172.24.24.134 TLSv1.3 89 Application Data
14162 292.004269100 172.24.24.134 104.18.32.47 TCP 54 64825 → 443 [ACK] Seq=421310 Ack=369423 Win=261888 Len=0
14163 292.004313300 172.24.24.134 104.18.32.47 TCP 54 64825 → 443 [ACK] Seq=421310 Ack=369501 Win=261888 Len=0
14164 292.007241700 172.24.24.134 104.18.32.47 TLSv1.3 89 Application Data
14165 292.081884200 104.18.32.47 172.24.24.134 TCP 58 443 → 64825 [ACK] Seq=369501 Ack=421345 Win=778240 Len=0
14166 292.177189400 104.18.32.47 172.24.24.134 TLSv1.3 419 Application Data
14167 292.177190700 104.18.32.47 172.24.24.134 TLSv1.3 113 Application Data
14168 292.177191500 104.18.32.47 172.24.24.134 TLSv1.3 89 Application Data
14169 292.177358500 172.24.24.134 104.18.32.47 TCP 54 64825 → 443 [ACK] Seq=421345 Ack=369917 Win=261632 Len=0
14170 292.233143600 172.24.24.134 104.18.32.47 TCP 54 64825 → 443 [ACK] Seq=421345 Ack=369948 Win=261376 Len=0
14195 292.952639100 172.24.24.134 35.186.224.35 TLSv1.2 82 Application Data
14196 293.017262500 35.186.224.35 172.24.24.134 TCP 54 443 → 53283 [ACK] Seq=1 Ack=281 Win=1050 Len=0
14200 293.099019700 172.24.24.134 35.186.224.35 TLSv1.2 97 Application Data
14201 293.123044900 35.186.224.35 172.24.24.134 TCP 54 443 → 51724 [ACK] Seq=361 Ack=431 Win=1043 Len=0
14202 293.221820500 35.186.224.35 172.24.24.134 TLSv1.2 94 Application Data
14203 293.273430400 172.24.24.134 35.186.224.35 TCP 54 51724 → 443 [ACK] Seq=431 Ack=401 Win=255 Len=0
Ethernet II, Src: Intel_9c:ba:6c (64:4a:7d:9c:ba:6c), Dst: Cisco_4e:e7:77 (00:42:5a:4e:e7:77)
Internet Protocol Version 4, Src: 172.24.24.134, Dst: 35.186.224.22
Transmission Control Protocol, Src Port: 65370, Dst Port: 443, Seq: 0, Len: 0
Source Port: 65370
Destination Port: 443
[Stream index: 0]
[Stream Packet Number: 1]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 2738940513
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 .... = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
Window: 65535
[Calculated window size: 65535]
0000 00 42 5a 4e e7 77 64 4a 7d 9c ba 6c 08 00 45 00
0010 00 34 d4 6b 40 00 80 06 00 00 ac 18 18 86 23 ba
0020 e0 16 ff 5a 01 bb a3 40 ea 61 00 00 00 00 80 02
0030 ff ff c8 95 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02
Source Port (tcp.srcport), 2 bytes
Packets: 14258 · Displayed: 3662 (25.7%)
Profile: Default

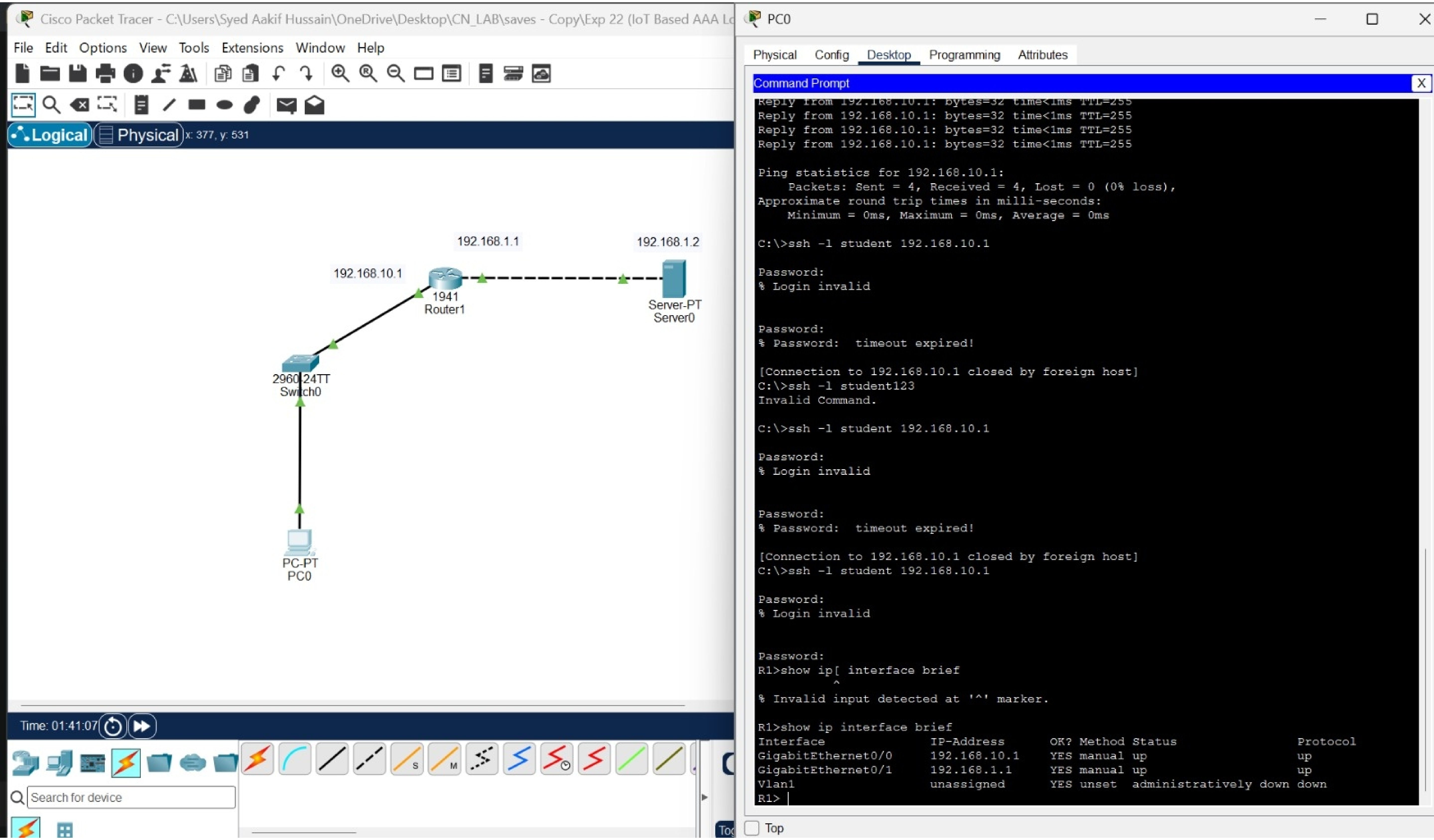


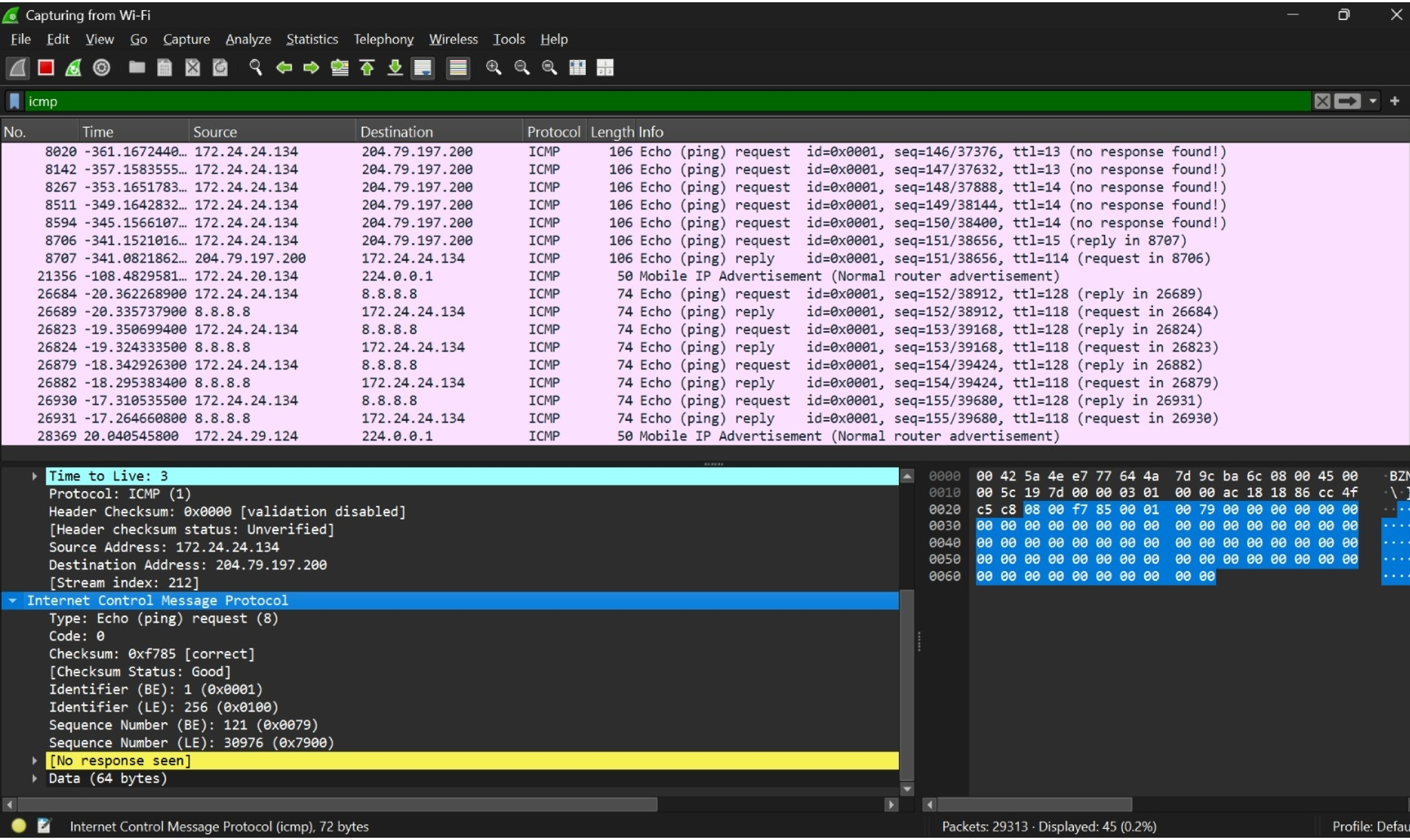
Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 735, y: 275
Root
PC-PT
PC0
2960-24TT
Switch0
Server-PT
Server0
PC-PT
PC1
PDU Information at Device: Switch0
OSI Model
Inbound PDU Details
Outbound PDU Details
PDU Formats
SRC IP:192.168.1.2
DST IP:192.168.1.10
DATA (VARIABLE LENGTH)
TCP
0 4 8 16 24 Bits
SOURCE PORT:1031
DESTINATION PORT:80
SEQUENCE NUMBER:0
ACKNOWLEDGEMENT NUMBER:2
OFFSE T:0x0
RESER VED: 0
FLAGS:0b00010 100
WINDOW:0
CHECKSUM:0x0000
URGENT POINTER:0x0000
OPTION
DATA (VARIABLE LENGTH)
PADDING: 0
Simulation Panel
Vis.
Time(sec)
Last Device
150.035 PC0
150.035 Switch0
150.035 Server0
150.035 Switch0
150.035 --
150.036 PC0
150.036 Switch0
150.036 Switch0
150.036 --
150.037 PC0
150.037 Switch0
150.037 --
150.038 PC0
150.038 Switch0
Visible 150.039 Switch0
Reset Simulation
Constant Delay
Captured to 1273.492 s
Edit Filters
Show All/None
Event List
Realtime
Simulation
Server0
URL http://192.168.1.10
Go
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunitie Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image
Time: 01:17:47.9
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete


Cisco Packet Tracer - C:\Users\sss\Cisco Packet Tracer 9.0.0\saves\How to connect Two different network using router in Csco Packet Tracer.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 658, y: 14
Root
00:56:30
How to connect Two different network using router in Csco Packet Tracer
110.1.1.3
ISR4331
Router0
192.168.1.3
LAN CONNECTION 1
LAN CONNECTION 2
Switch0
Switch
PC-PT
Computer 1
PC-PT
Computer 2
PC-PT
Computer 3
PC-PT
Computer 4
10.1.1.1
10.1.1.2
192.168.1.1
192.168.1.2
Time: 00:01:51
Realtime
Simulation
PT8200
4331
4321
IR8340
1941
2901
2911
819IOX
819HGW
829
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Comp...
Computer 4
ICMP
0.000
N
0
(e...


Server-PT
Server0
PC-PT
PC0
Switch0
PC-PT
PC2
PC-PT
PC1


PC-PT
PC0
2960-24TT
Switch0
PC-PT
PC1
2960-24TT
Switch1
PC-PT
PC2
2960-24TT
Switch2
PC-PT
PC3
Switch3
PC-PT
PC4
2960-24TT
Switch4
PC-PT
PC5
2960-24TT
Switch5
PC-PT
PC6
PC-PT
PC7


Cisco Packet Tracer - C:\Users\Syed Aakif Hussain\OneDrive\Desktop\CN_LAB\saves - Copy\Exp 22 (IoT Based AAA Lo
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 377, y: 531
192.168.1.1
192.168.1.2
192.168.10.1
1941
Router1
Server-PT
Server0
2960-24TT
Switch0
PC-PT
PC0
Time: 01:41:07
Search for device
PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ssh -l student 192.168.10.1
Password:
% Login invalid
Password:
% Password: timeout expired!
[Connection to 192.168.10.1 closed by foreign host]
C:\>ssh -l student123
Invalid Command.
C:\>ssh -l student 192.168.10.1
Password:
% Login invalid
Password:
% Password: timeout expired!
[Connection to 192.168.10.1 closed by foreign host]
C:\>ssh -l student 192.168.10.1
Password:
% Login invalid
Password:
R1>show ip[interface brief
^
% Invalid input detected at '^' marker.
R1>show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.10.1 YES manual up up
GigabitEthernet0/1 192.168.1.1 YES manual up up
Vlan1 unassigned YES unset administratively down down
R1>
Top



Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
8020 -361.1672440… 172.24.24.134 204.79.197.200 ICMP 106 Echo (ping) request id=0x0001, seq=146/37376, ttl=13 (no response found!)
8142 -357.1583555… 172.24.24.134 204.79.197.200 ICMP 106 Echo (ping) request id=0x0001, seq=147/37632, ttl=13 (no response found!)
8267 -353.1651783… 172.24.24.134 204.79.197.200 ICMP 106 Echo (ping) request id=0x0001, seq=148/37888, ttl=14 (no response found!)
8511 -349.1642832… 172.24.24.134 204.79.197.200 ICMP 106 Echo (ping) request id=0x0001, seq=149/38144, ttl=14 (no response found!)
8594 -345.1566107… 172.24.24.134 204.79.197.200 ICMP 106 Echo (ping) request id=0x0001, seq=150/38400, ttl=14 (no response found!)
8706 -341.1521016… 172.24.24.134 204.79.197.200 ICMP 106 Echo (ping) request id=0x0001, seq=151/38656, ttl=15 (reply in 8707)
8707 -341.0821862… 204.79.197.200 172.24.24.134 ICMP 106 Echo (ping) reply id=0x0001, seq=151/38656, ttl=114 (request in 8706)
21356 -108.4829581… 172.24.20.134 224.0.0.1 ICMP 50 Mobile IP Advertisement (Normal router advertisement)
26684 -20.362268900 172.24.24.134 8.8.8.8 ICMP 74 Echo (ping) request id=0x0001, seq=152/38912, ttl=128 (reply in 26689)
26689 -20.335737900 8.8.8.8 172.24.24.134 ICMP 74 Echo (ping) reply id=0x0001, seq=152/38912, ttl=118 (request in 26684)
26823 -19.350699400 172.24.24.134 8.8.8.8 ICMP 74 Echo (ping) request id=0x0001, seq=153/39168, ttl=128 (reply in 26824)
26824 -19.324333500 8.8.8.8 172.24.24.134 ICMP 74 Echo (ping) reply id=0x0001, seq=153/39168, ttl=118 (request in 26823)
26879 -18.342926300 172.24.24.134 8.8.8.8 ICMP 74 Echo (ping) request id=0x0001, seq=154/39424, ttl=128 (reply in 26882)
26882 -18.295383400 8.8.8.8 172.24.24.134 ICMP 74 Echo (ping) reply id=0x0001, seq=154/39424, ttl=118 (request in 26879)
26930 -17.310535500 172.24.24.134 8.8.8.8 ICMP 74 Echo (ping) request id=0x0001, seq=155/39680, ttl=128 (reply in 26931)
26931 -17.264660800 8.8.8.8 172.24.24.134 ICMP 74 Echo (ping) reply id=0x0001, seq=155/39680, ttl=118 (request in 26930)
28369 20.040545800 172.24.29.124 224.0.0.1 ICMP 50 Mobile IP Advertisement (Normal router advertisement)
Time to Live: 3
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 172.24.24.134
Destination Address: 204.79.197.200
[Stream index: 212]
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0xf785 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 121 (0x0079)
Sequence Number (LE): 30976 (0x7900)
[No response seen]
Data (64 bytes)
0000 00 42 5a 4e e7 77 64 4a 7d 9c ba 6c 08 00 45 00
0010 00 5c 19 7d 00 00 03 01 00 00 ac 18 18 86 cc 4f
0020 c5 c8 08 00 f7 85 00 01 00 79 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00
Internet Control Message Protocol (icmp), 72 bytes
Packets: 29313 · Displayed: 45 (0.2%)
Profile: Default

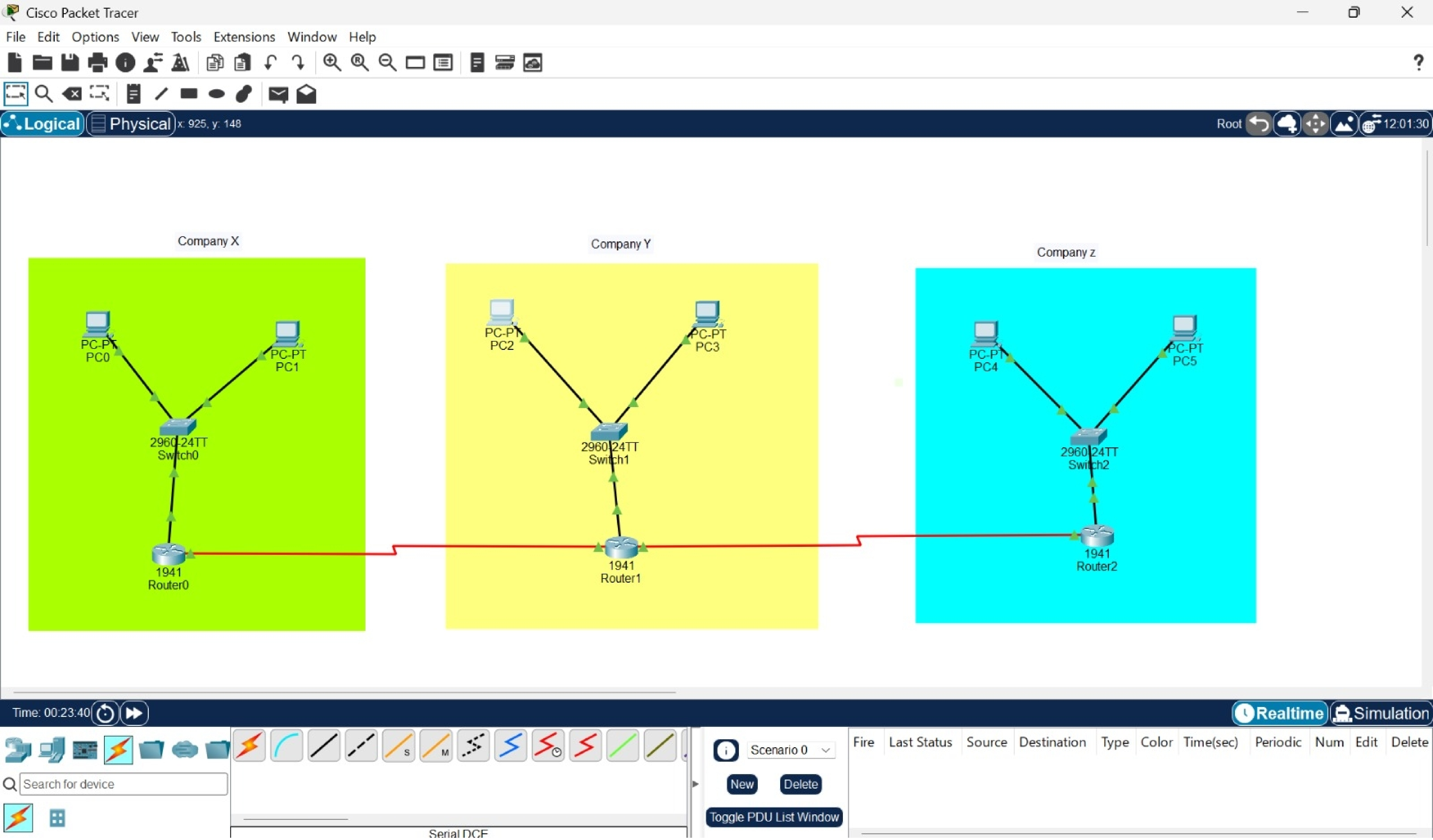
Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <string.h>

int main() {

    char hostname[100];
    char ip[100];

    printf("====================================\n");
    printf("   DNS SERVER - CLIENT SIMULATION\n");
    printf("====================================\n");

    printf("\nEnter hostname: ");
    scanf("%s", hostname);

    // DNS Server lookup simulation
    if (strcmp(hostname, "www.google.com") == 0) {
        strcpy(ip, "142.250.195.100");
    }
    else if (strcmp(hostname, "www.yahoo.com") == 0) {
        strcpy(ip, "98.137.11.163");
    }
    else if (strcmp(hostname, "www.example.com") == 0) {
        strcpy(ip, "93.184.216.34");
```

input

```
IP Address : 142.250.195.100

DNS simulation completed successfully.


...Program finished with exit code 0
Press ENTER to exit console.
```


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 925, y: 148
Root
12:01:30
Company X
Company Y
Company z
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
2960-24TT
Switch0
2960-24TT
Switch1
2960-24TT
Switch2
1941
Router0
1941
Router1
1941
Router2
Time: 00:23:40
Realtime
Simulation
Search for device
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Serial DCE

Run Debug Stop Share Save {} Beautify Language C

main.c

```c
#include <stdio.h>

int main() {
    int frames, window, i, j;

    printf("Enter number of frames: ");
    scanf("%d", &frames);

    printf("Enter window size: ");
    scanf("%d", &window);

    for (i = 1; i <= frames; i += window) {

        printf("\nSending frames:\n");

        for (j = i; j < i + window && j <= frames; j++) {
            printf("Frame %d sent\n", j);
        }

        printf("Acknowledgement received for frames ");

```

input

```
Enter number of frames: 8
Enter window size: 3

Sending frames:
Frame 1 sent
Frame 2 sent
Frame 3 sent
Acknowledgement received for frames 1 2 3
Window slides forward.
```

Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <netdb.h>
#include <arpa/inet.h>

int main() {

    char hostname[100];
    struct hostent *host;
    struct in_addr **address_list;

    printf("Enter hostname: ");
    scanf("%99s", hostname);

    // Resolve hostname
    host = gethostbyname(hostname);

    if (host == NULL) {
        printf("DNS resolution failed.\n");
        return 1;
    }

    printf("\nHostname: %s\n", hostname);
    printf("IP Address(es):\n");
```

input

```
Enter hostname: www.google.com

Hostname: www.google.com
IP Address(es):
142.251.157.119
142.251.153.119
142.251.150.119
```

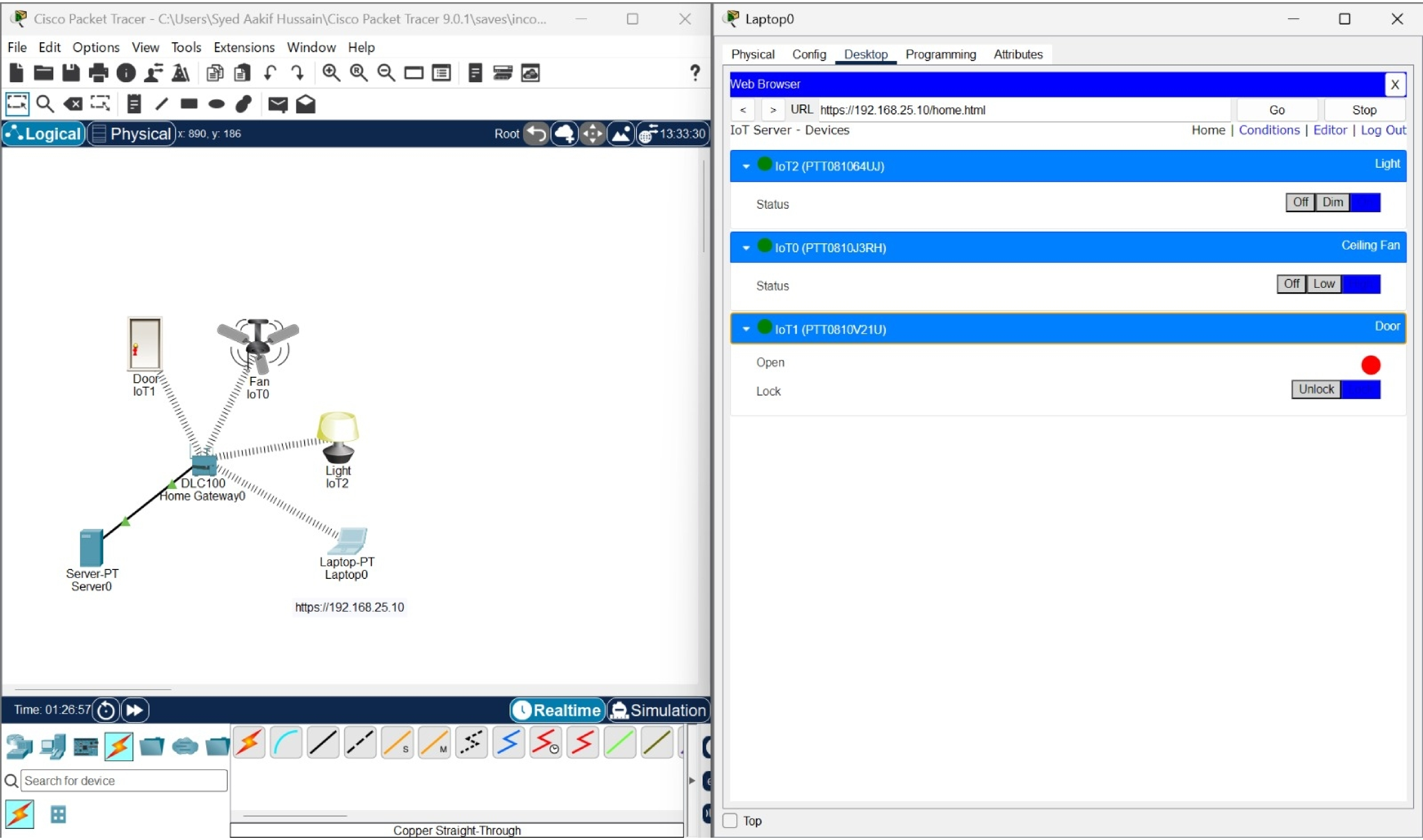
Run Debug Stop Share Save { } Beautify

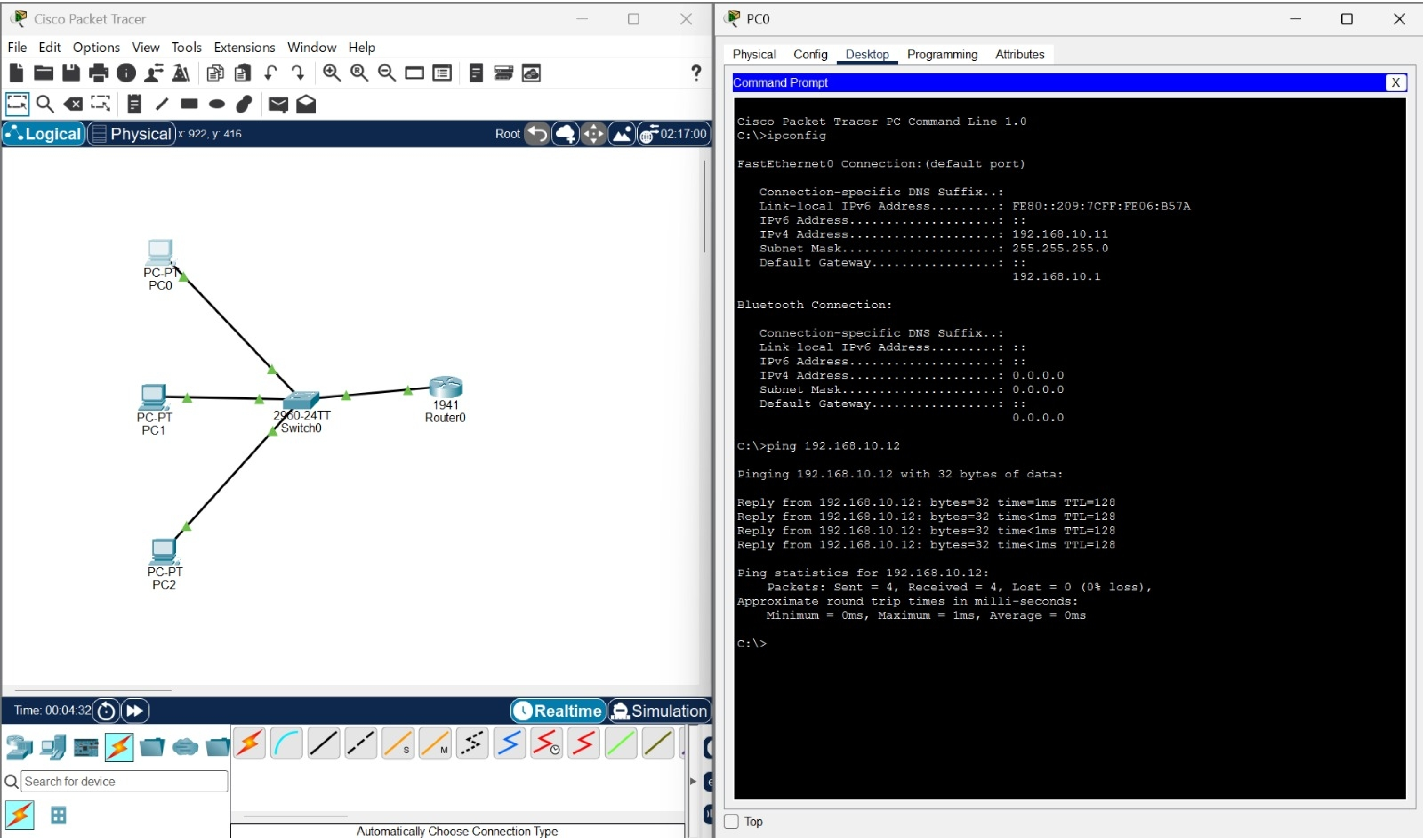
main.c

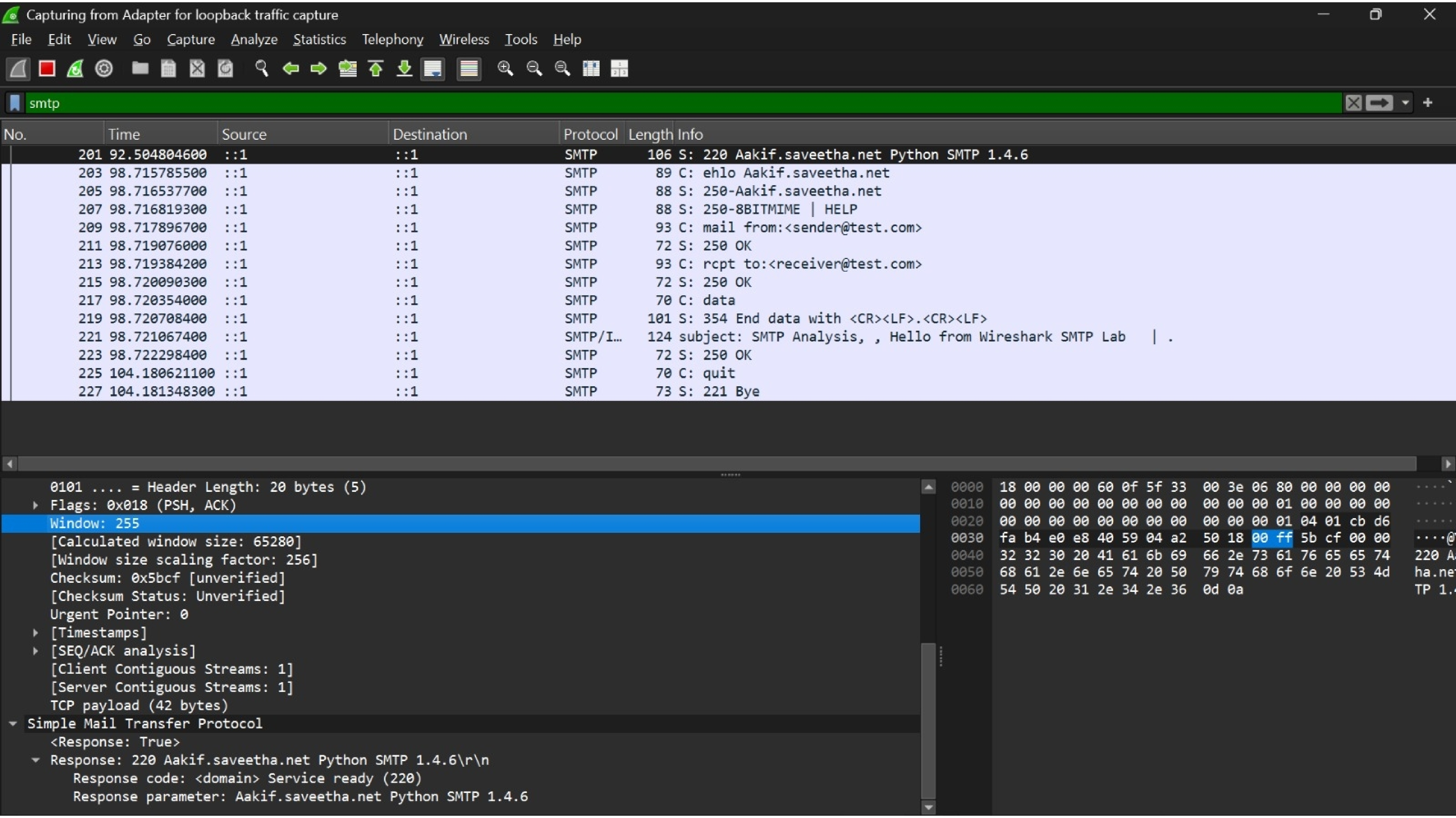
```c
#include <stdio.h>
#include <string.h>

int main() {

    char ip[3][20] = {
        "192.168.1.1",
        "192.168.1.2",
        "192.168.1.5"
    };

    char mac[3][20] = {
        "AA:AA:AA:AA:AA:01",
        "AA:AA:AA:AA:AA:02",
        "BB:BB:BB:BB:BB:05"
    };

    char searchIP[20];
    int found = 0;

    printf("Enter IP address: ");
```

input

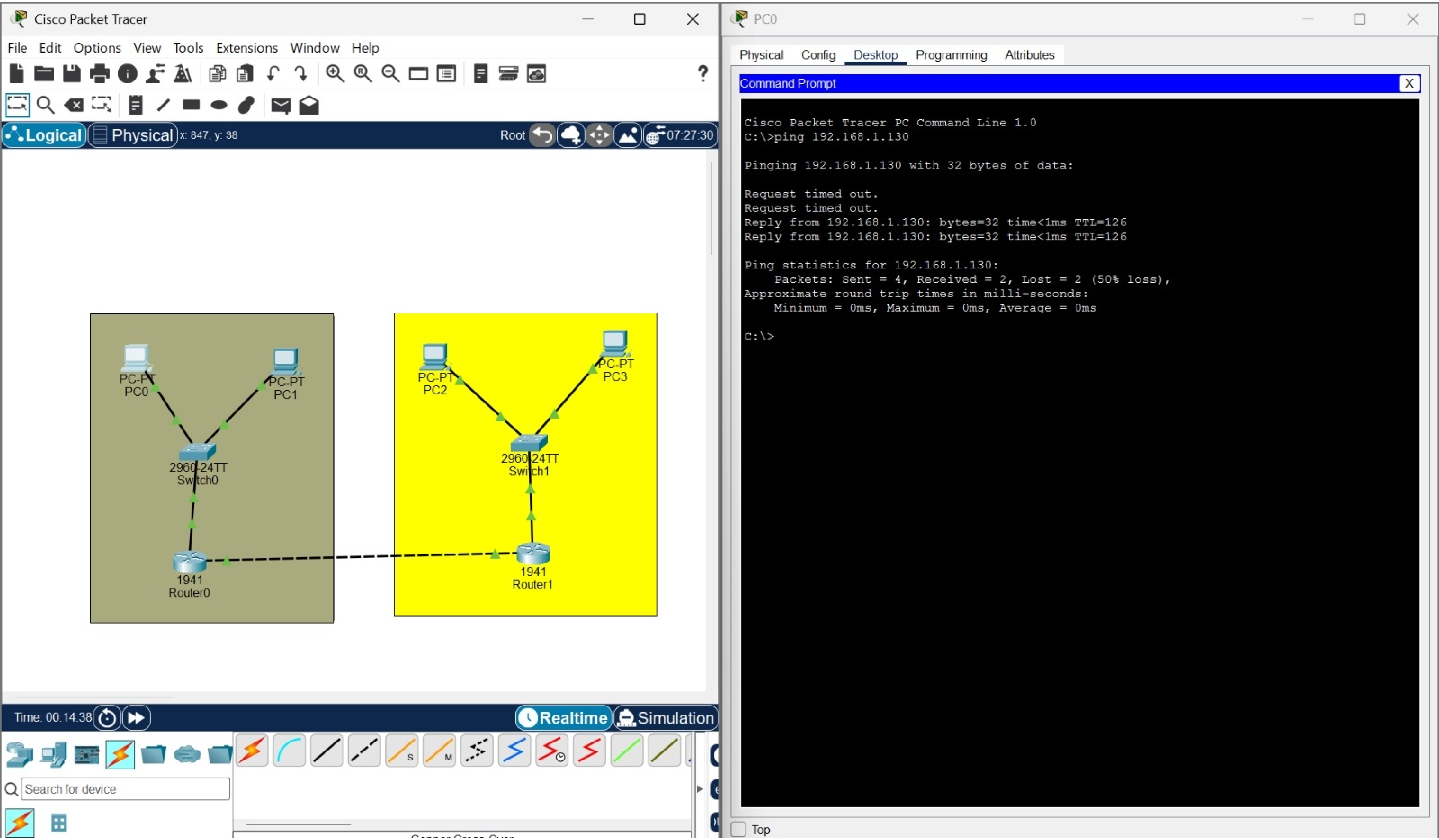
```
Enter IP address: 192.168.1.5
IP Address  : 192.168.1.5
MAC Address : BB:BB:BB:BB:BB:05


...Program finished with exit code 0
Press ENTER to exit console.
```

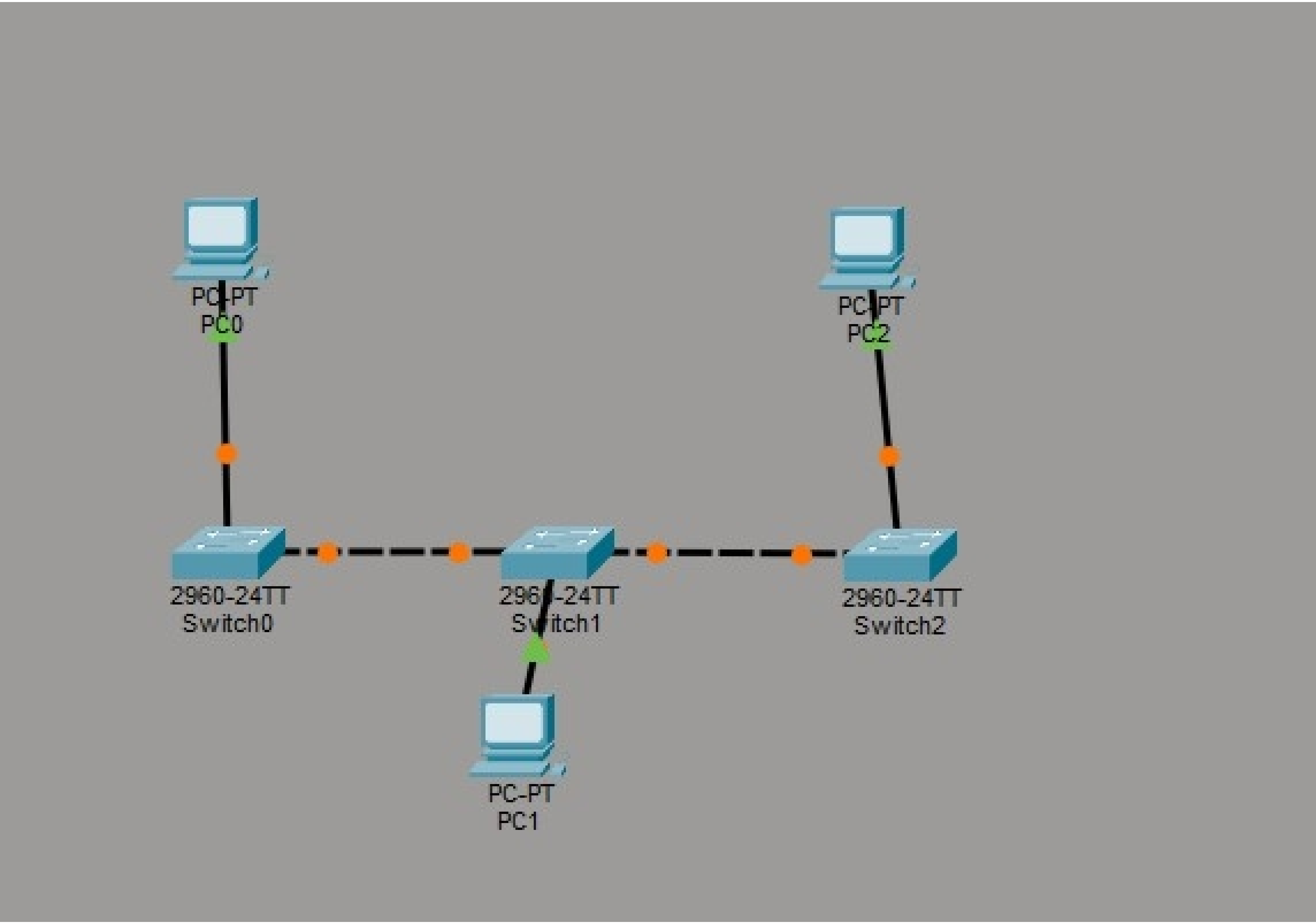

Cisco Packet Tracer - C:\Users\Syed Aakif Hussain\Cisco Packet Tracer 9.0.1\saves\inco...
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 890, y: 186
Root
13:33:30
Door
IoT1
Fan
IoT0
Light
IoT2
DLC100
Home Gateway0
Server-PT
Server0
Laptop-PT
Laptop0
https://192.168.25.10
Time: 01:26:57
Realtime
Simulation
Search for device
Copper Straight-Through
Laptop0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
URL https://192.168.25.10/home.html
Go
Stop
IoT Server - Devices
Home | Conditions | Editor | Log Out
IoT2 (PTT081064UJ)
Light
Status
Off
Dim
IoT0 (PTT0810J3RH)
Ceiling Fan
Status
Off
Low
IoT1 (PTT0810V21U)
Door
Open
Lock
Unlock
Top


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 922, y: 416
Root
02:17:00
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
2960-24TT
Switch0
1941
Router0
Time: 00:04:32
Realtime
Simulation
Search for device
Automatically Choose Connection Type
PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: FE80::209:7CFF:FE06:B57A
IPv6 Address....................: ::
IPv4 Address....................: 192.168.10.11
Subnet Mask.....................: 255.255.255.0
Default Gateway.................: ::
192.168.10.1
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: ::
IPv6 Address....................: ::
IPv4 Address....................: 0.0.0.0
Subnet Mask.....................: 0.0.0.0
Default Gateway.................: ::
0.0.0.0
C:\>ping 192.168.10.12
Pinging 192.168.10.12 with 32 bytes of data:
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

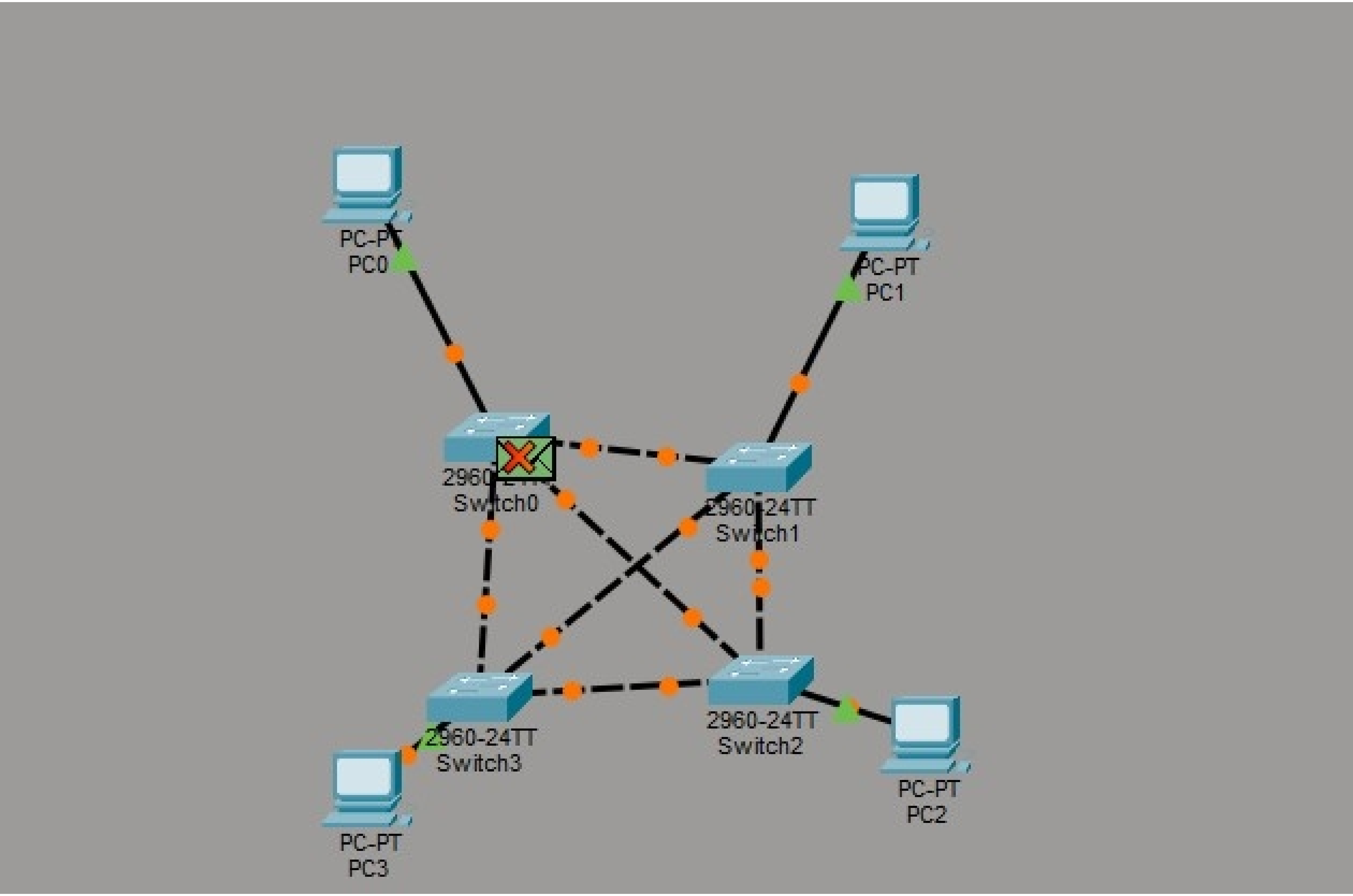


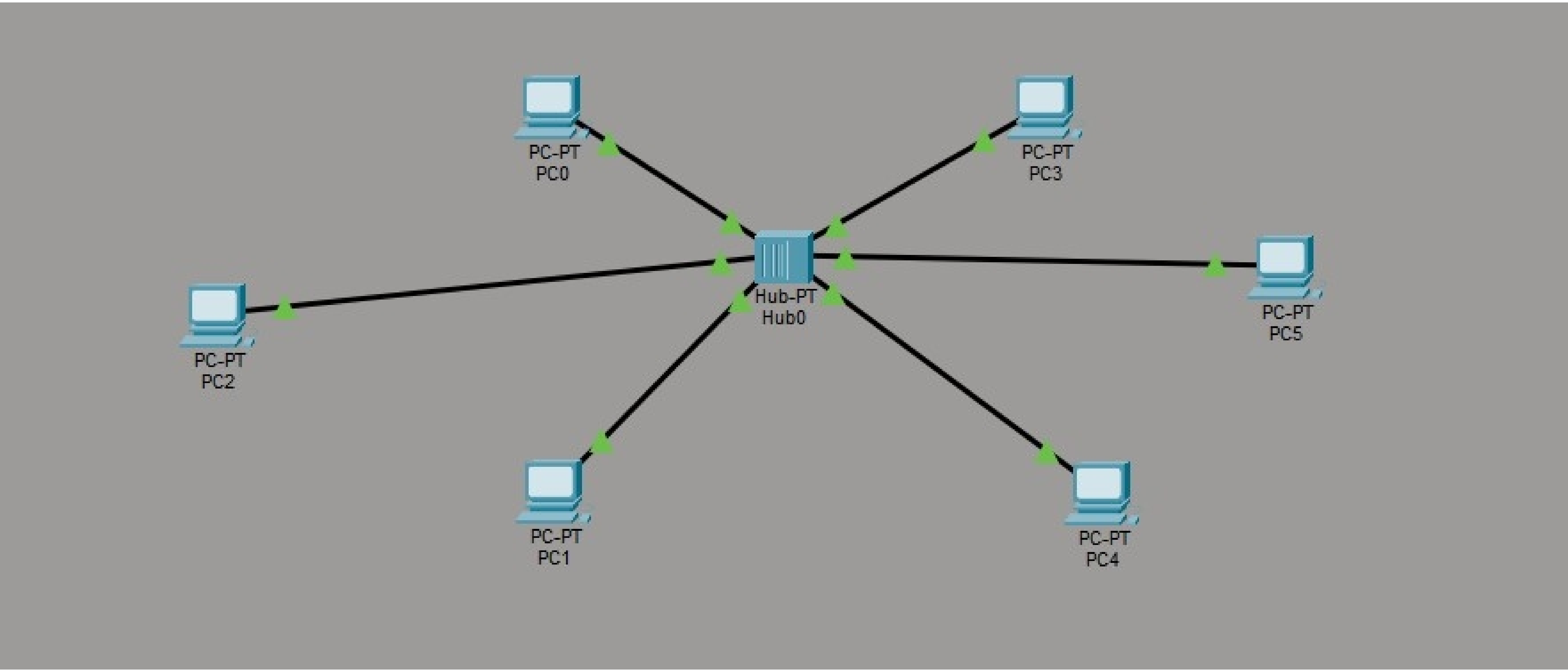
Capturing from Adapter for loopback traffic capture
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
smtp
No. Time Source Destination Protocol Length Info
201 92.504804600 ::1 ::1 SMTP 106 S: 220 Aakif.saveetha.net Python SMTP 1.4.6
203 98.715785500 ::1 ::1 SMTP 89 C: ehlo Aakif.saveetha.net
205 98.716537700 ::1 ::1 SMTP 88 S: 250-Aakif.saveetha.net
207 98.716819300 ::1 ::1 SMTP 88 S: 250-8BITMIME | HELP
209 98.717896700 ::1 ::1 SMTP 93 C: mail from:<sender@test.com>
211 98.719076000 ::1 ::1 SMTP 72 S: 250 OK
213 98.719384200 ::1 ::1 SMTP 93 C: rcpt to:<receiver@test.com>
215 98.720090300 ::1 ::1 SMTP 72 S: 250 OK
217 98.720354000 ::1 ::1 SMTP 70 C: data
219 98.720708400 ::1 ::1 SMTP 101 S: 354 End data with <CR><LF>.<CR><LF>
221 98.721067400 ::1 ::1 SMTP/I… 124 subject: SMTP Analysis, , Hello from Wireshark SMTP Lab | .
223 98.722298400 ::1 ::1 SMTP 72 S: 250 OK
225 104.180621100 ::1 ::1 SMTP 70 C: quit
227 104.181348300 ::1 ::1 SMTP 73 S: 221 Bye
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x5bcf [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (42 bytes)
Simple Mail Transfer Protocol
<Response: True>
Response: 220 Aakif.saveetha.net Python SMTP 1.4.6\r\n
Response code: <domain> Service ready (220)
Response parameter: Aakif.saveetha.net Python SMTP 1.4.6
0000 18 00 00 00 60 0f 5f 33 00 3e 06 80 00 00 00 00
0010 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00
0020 00 00 00 00 00 00 00 00 00 00 00 01 04 01 cb d6
0030 fa b4 e0 e8 40 59 04 a2 50 18 00 ff 5b cf 00 00
0040 32 32 30 20 41 61 6b 69 66 2e 73 61 76 65 65 74
0050 68 61 2e 6e 65 74 20 50 79 74 68 6f 6e 20 53 4d
0060 54 50 20 31 2e 34 2e 36 0d 0a

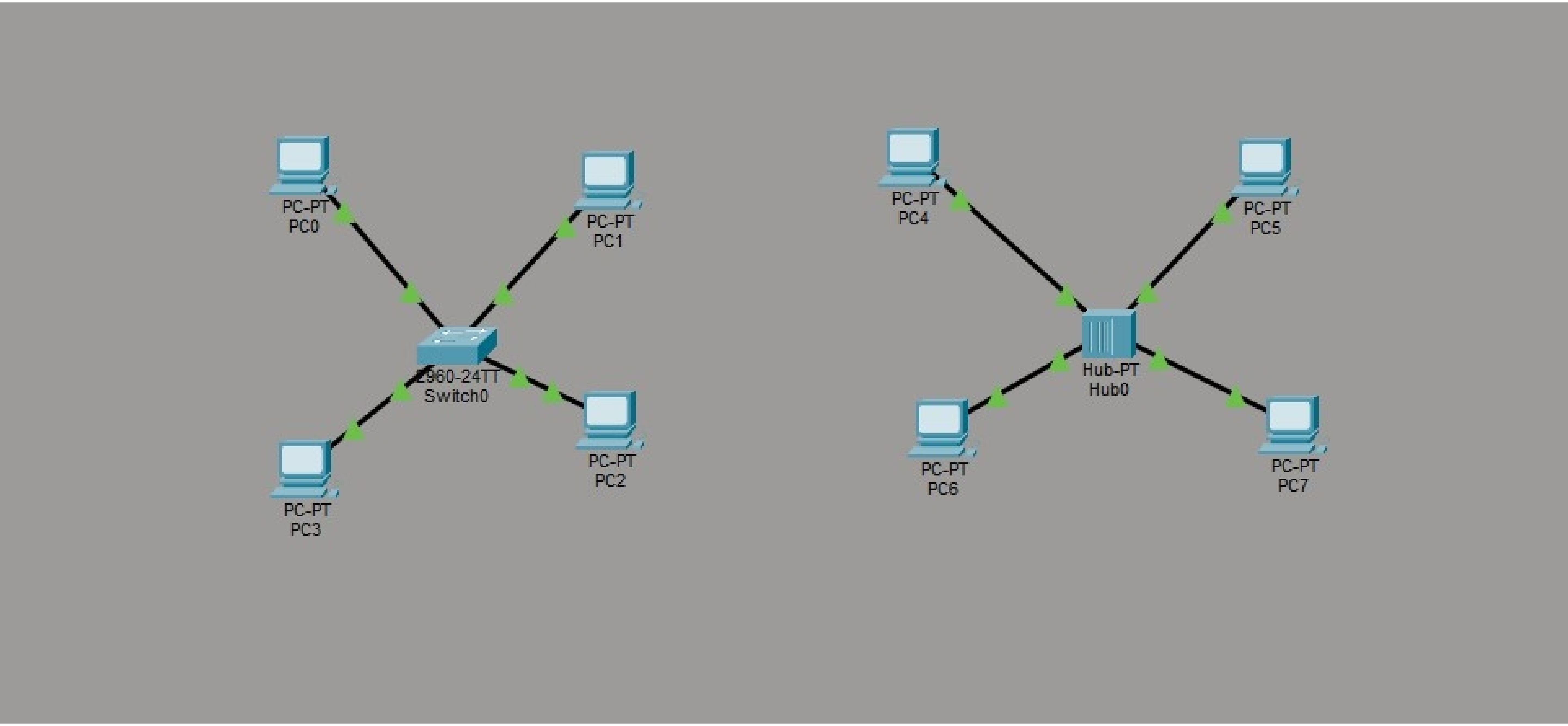

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 847, y: 38
Root
07:27:30
PC-PT
PC0
PC-PT
PC1
2960-24TT
Switch0
1941
Router0
PC-PT
PC2
PC-PT
PC3
2960-24TT
Switch1
1941
Router1
Time: 00:14:38
Realtime
Simulation
Search for device
PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.130
Pinging 192.168.1.130 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.1.130: bytes=32 time<1ms TTL=126
Reply from 192.168.1.130: bytes=32 time<1ms TTL=126
Ping statistics for 192.168.1.130:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top

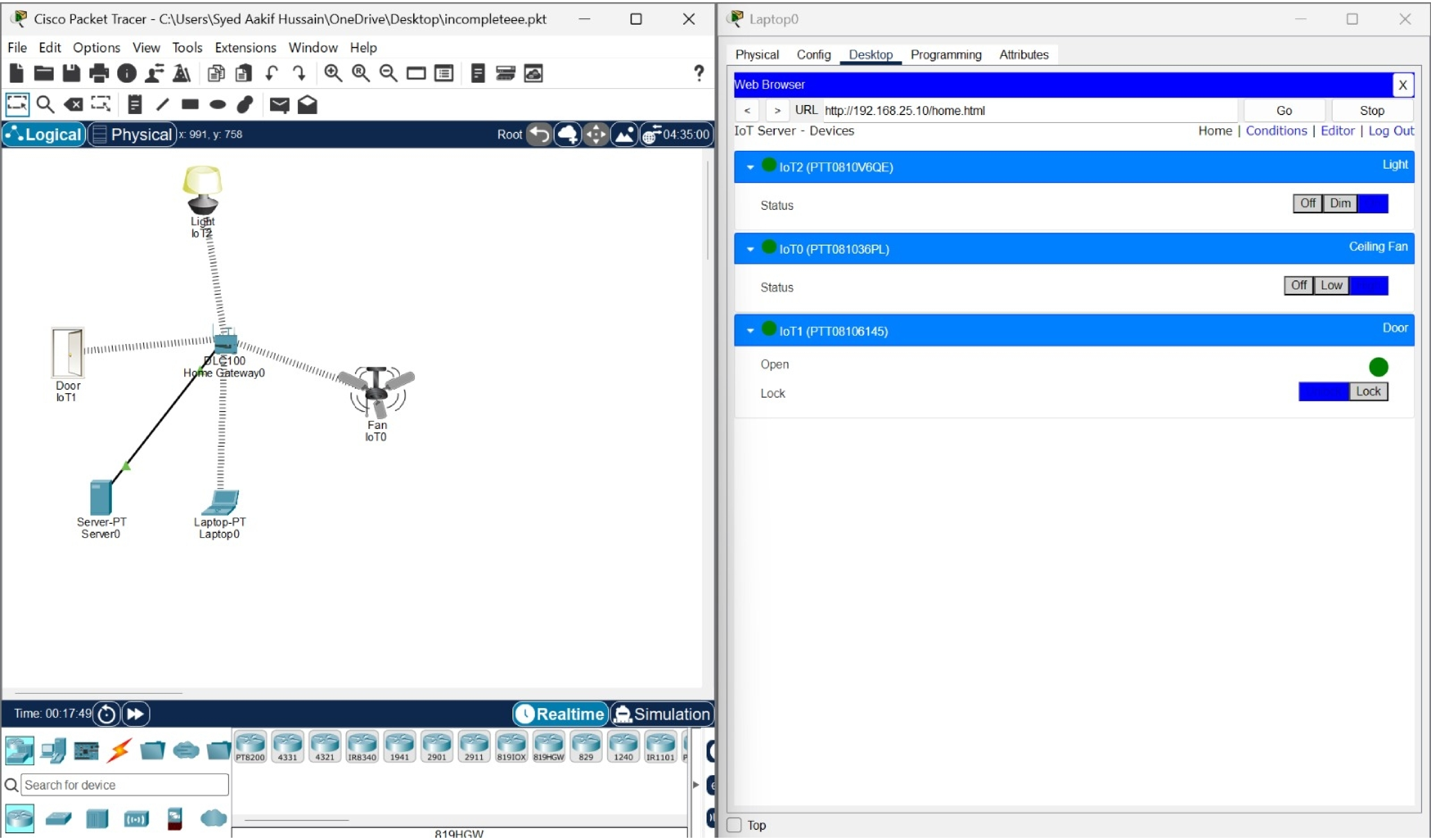


PC-PT
PC0
PC-PT
PC2
2960-24TT
Switch0
2960-24TT
Switch1
2960-24TT
Switch2
PC-PT
PC1


PC-PT
PC0
PC-PT
PC1
2960-24TT
Switch0
2960-24TT
Switch1
2960-24TT
Switch2
2960-24TT
Switch3
PC-PT
PC2
PC-PT
PC3


PC-PT
PC0
PC-PT
PC3
PC-PT
PC2
Hub-PT
Hub0
PC-PT
PC5
PC-PT
PC1
PC-PT
PC4


PC-PT
PC0
PC-PT
PC1
2960-24TT
Switch0
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
Hub-PT
Hub0
PC-PT
PC6
PC-PT
PC7


Cisco Packet Tracer - C:\Users\Syed Aakif Hussain\OneDrive\Desktop\incompleteee.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 991, y: 758
Root
04:35:00
Light
IoT2
DLC100
Home Gateway0
Door
IoT1
Fan
IoT0
Server-PT
Server0
Laptop-PT
Laptop0
Time: 00:17:49
Realtime
Simulation
Search for device
819HGW
Laptop0
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL http://192.168.25.10/home.html
Go
Stop
IoT Server - Devices
Home | Conditions | Editor | Log Out
IoT2 (PTT0810V6QE)
Light
Status
Off
Dim
IoT0 (PTT081036PL)
Ceiling Fan
Status
Off
Low
IoT1 (PTT08106145)
Door
Open
Lock
Lock
Top

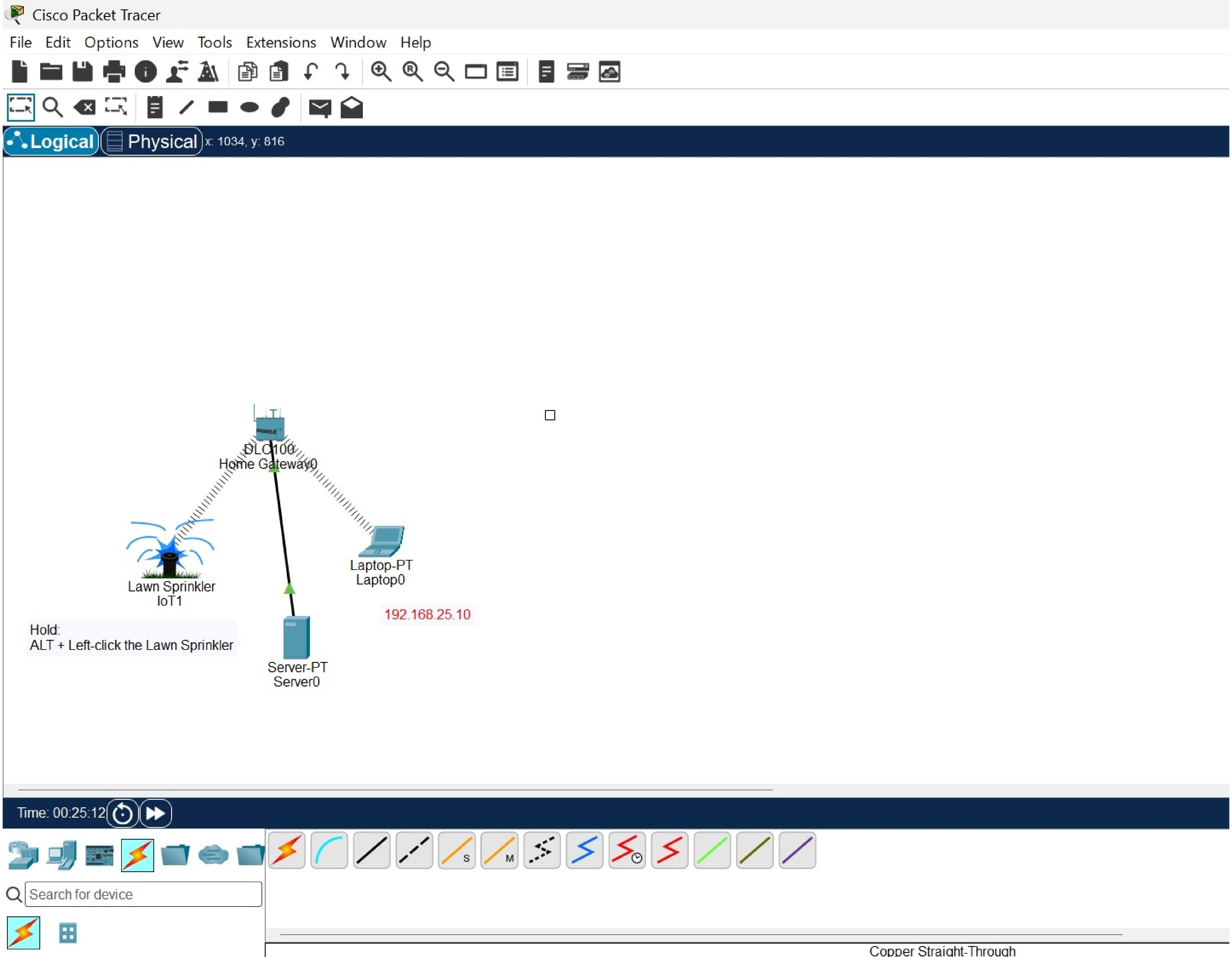
Run Debug Stop Share Save { } Beautify

main.c

```c
    printf("    TCP DATE AND TIME APPLICATION\n");
    printf("====================================\n");

    // Simulating client connection
    printf("\nClient connecting to server...\n");
    printf("Server accepted the connection.\n");

    // Server gets current date and time
    currentTime = time(NULL);
    dateTime = ctime(&currentTime);

    printf("\nServer gets current Date and Time.\n");

    // Server sends date and time
    printf("Server sends: %s", dateTime);

    // Client receives date and time
    printf("\nClient received Date and Time:\n");
    printf("%s", dateTime);

    printf("\nConnection closed successfully.\n");

    return 0;
}
```

input

```
Server gets current Date and Time.
Server sends: Sat Aug 29 09:25:33 2026

Client received Date and Time:
Sat Aug 29 09:25:33 2026

Connection closed successfully.
```


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1034, y: 816
DLC100
Home Gateway0
Lawn Sprinkler
IoT1
Laptop-PT
Laptop0
192.168.25.10
Hold:
ALT + Left-click the Lawn Sprinkler
Server-PT
Server0
Time: 00:25:12
Search for device
Copper Straight-Through

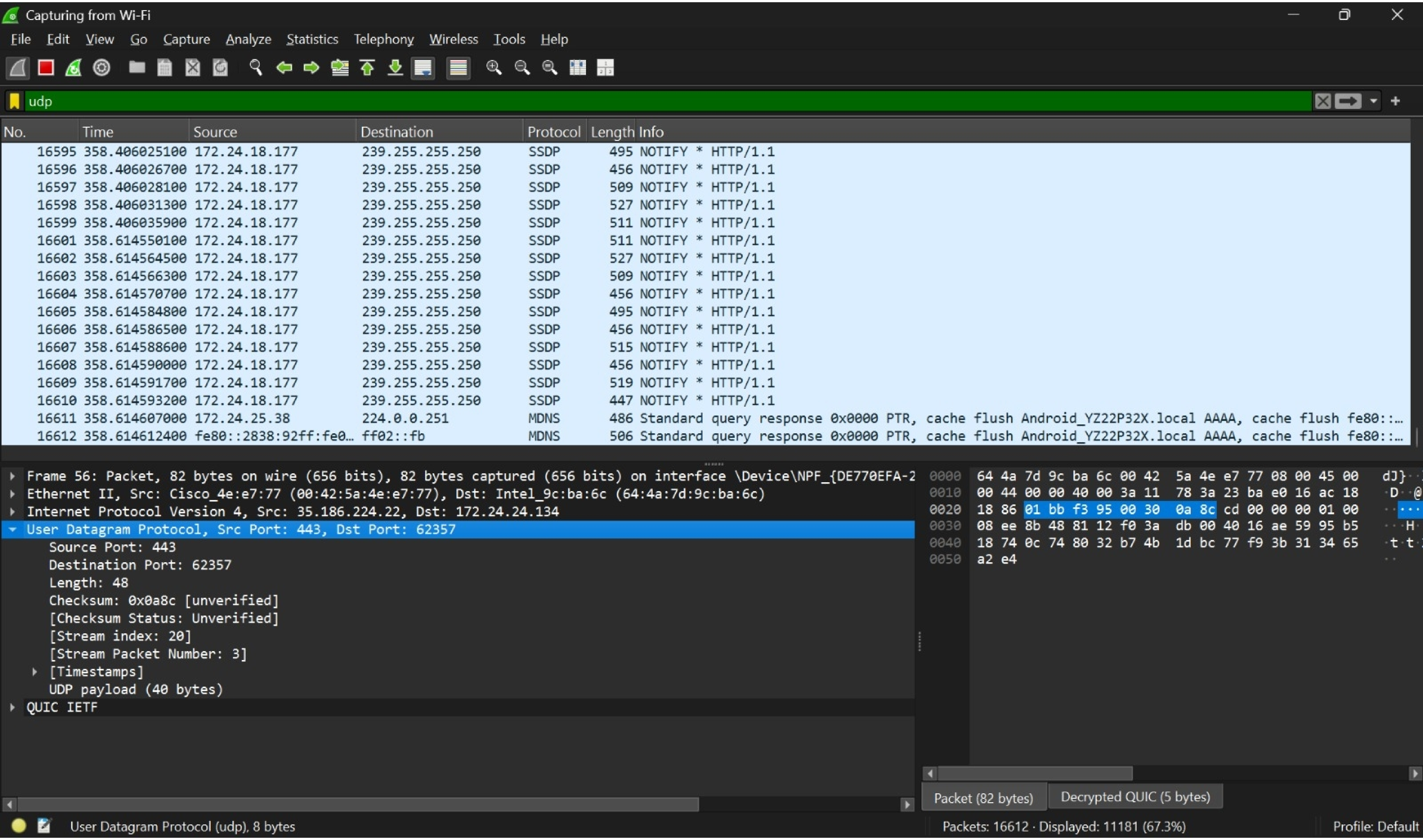
Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <string.h>

void xorOperation(char *data, char *divisor, int start) {
    int i;
    int divisorLength = strlen(divisor);

    for (i = 0; i < divisorLength; i++) {
        data[start + i] =
            (data[start + i] == divisor[i]) ? '0' : '1';
    }
}

int main() {
    char data[100], divisor[50], temp[150];
    int dataLength, divisorLength;
    int i;

    printf("Enter data: ");
    scanf("%s", data);

```

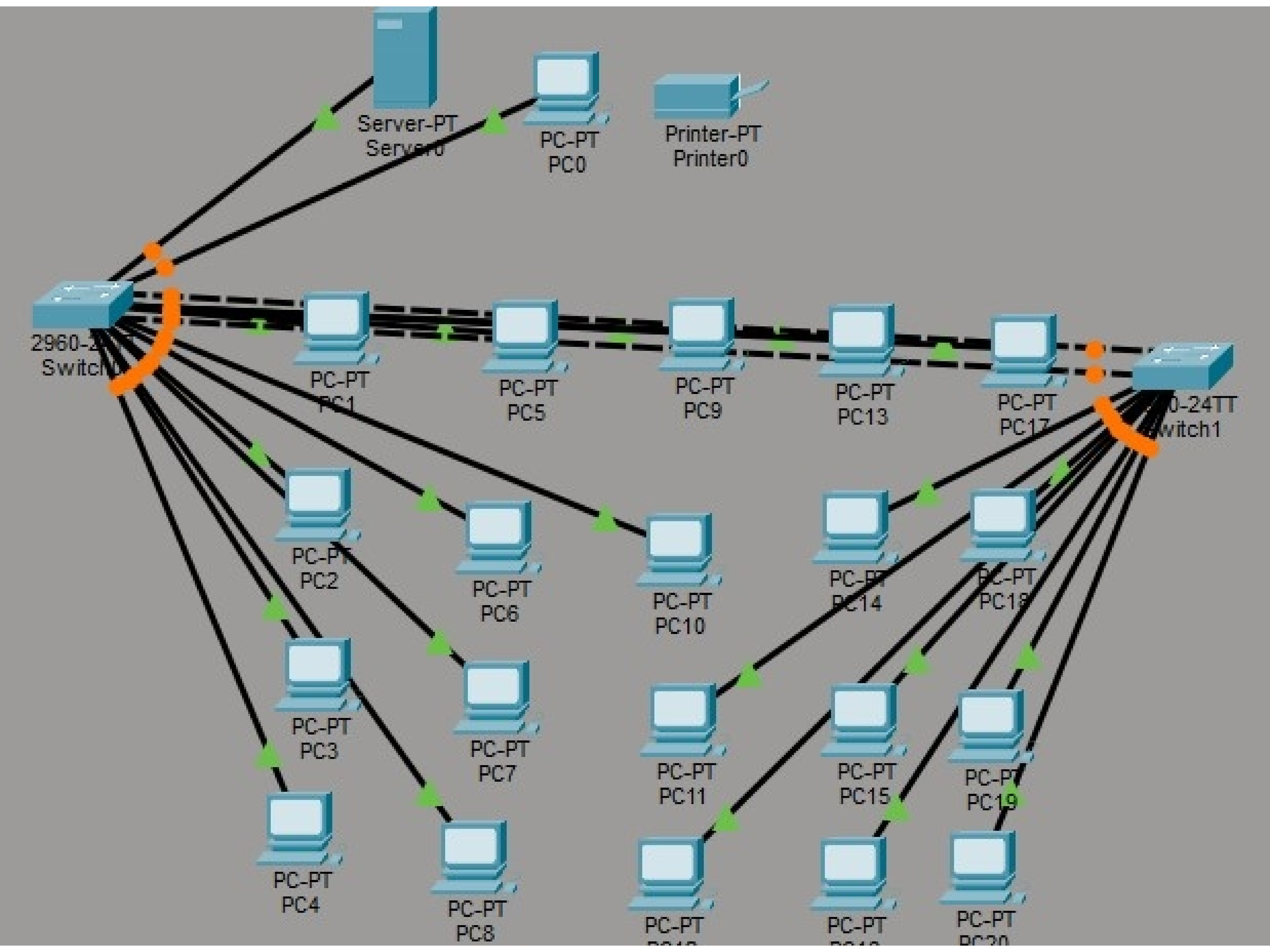
input

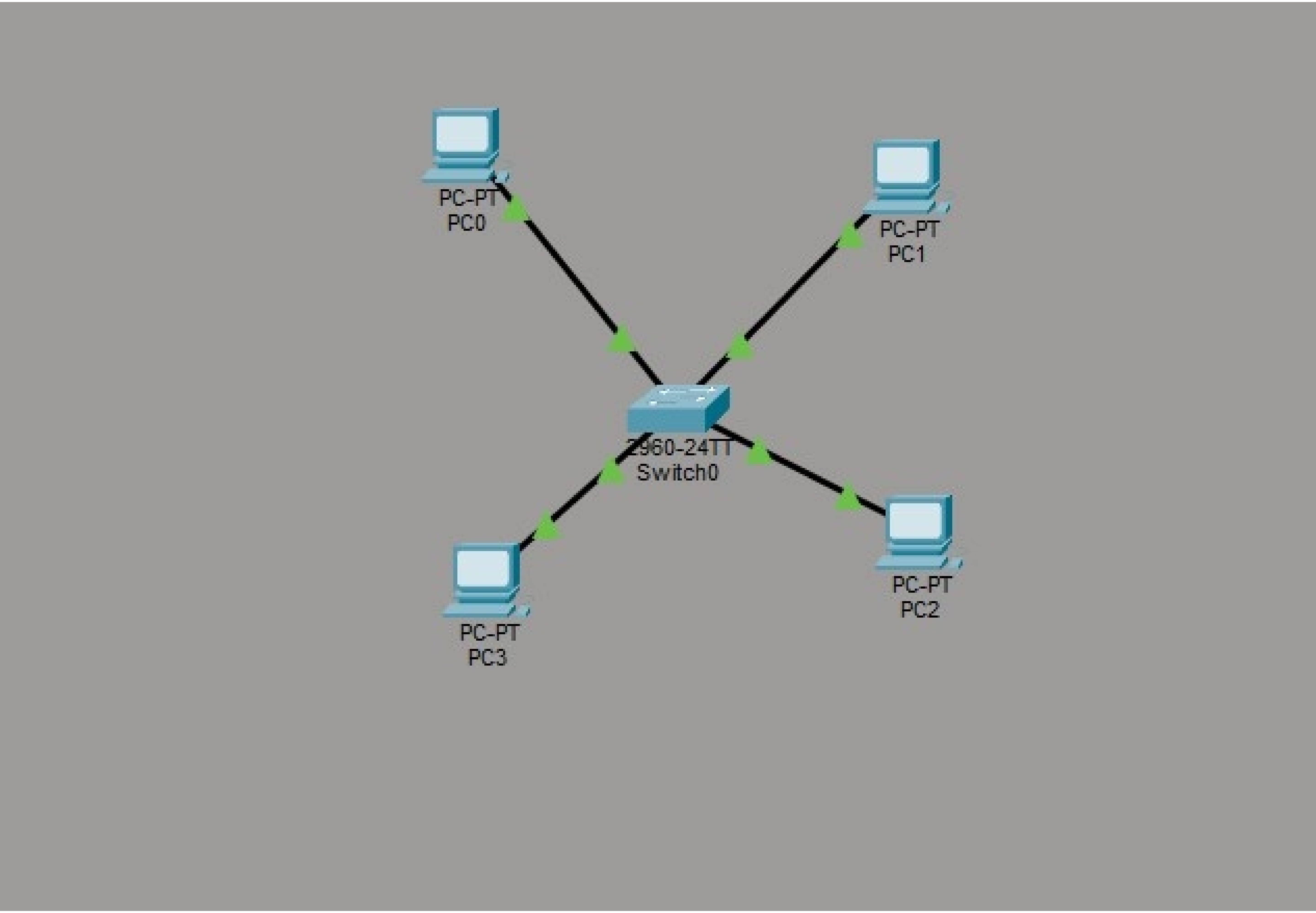
```
Enter data: 1101
Enter divisor: 1011

Data after appending zeros: 1101000
CRC bits: 001
Transmitted Codeword: 1101001


...Program finished with exit code 0
Press ENTER to exit console.
```



Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
16595 358.406025100 172.24.18.177 239.255.255.250 SSDP 495 NOTIFY * HTTP/1.1
16596 358.406026700 172.24.18.177 239.255.255.250 SSDP 456 NOTIFY * HTTP/1.1
16597 358.406028100 172.24.18.177 239.255.255.250 SSDP 509 NOTIFY * HTTP/1.1
16598 358.406031300 172.24.18.177 239.255.255.250 SSDP 527 NOTIFY * HTTP/1.1
16599 358.406035900 172.24.18.177 239.255.255.250 SSDP 511 NOTIFY * HTTP/1.1
16601 358.614550100 172.24.18.177 239.255.255.250 SSDP 511 NOTIFY * HTTP/1.1
16602 358.614564500 172.24.18.177 239.255.255.250 SSDP 527 NOTIFY * HTTP/1.1
16603 358.614566300 172.24.18.177 239.255.255.250 SSDP 509 NOTIFY * HTTP/1.1
16604 358.614570700 172.24.18.177 239.255.255.250 SSDP 456 NOTIFY * HTTP/1.1
16605 358.614584800 172.24.18.177 239.255.255.250 SSDP 495 NOTIFY * HTTP/1.1
16606 358.614586500 172.24.18.177 239.255.255.250 SSDP 456 NOTIFY * HTTP/1.1
16607 358.614588600 172.24.18.177 239.255.255.250 SSDP 515 NOTIFY * HTTP/1.1
16608 358.614590000 172.24.18.177 239.255.255.250 SSDP 456 NOTIFY * HTTP/1.1
16609 358.614591700 172.24.18.177 239.255.255.250 SSDP 519 NOTIFY * HTTP/1.1
16610 358.614593200 172.24.18.177 239.255.255.250 SSDP 447 NOTIFY * HTTP/1.1
16611 358.614607000 172.24.25.38 224.0.0.251 MDNS 486 Standard query response 0x0000 PTR, cache flush Android_YZ22P32X.local AAAA, cache flush fe80::…
16612 358.614612400 fe80::2838:92ff:fe0… ff02::fb MDNS 506 Standard query response 0x0000 PTR, cache flush Android_YZ22P32X.local AAAA, cache flush fe80::…
Frame 56: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{DE770EFA-2
Ethernet II, Src: Cisco_4e:e7:77 (00:42:5a:4e:e7:77), Dst: Intel_9c:ba:6c (64:4a:7d:9c:ba:6c)
Internet Protocol Version 4, Src: 35.186.224.22, Dst: 172.24.24.134
User Datagram Protocol, Src Port: 443, Dst Port: 62357
Source Port: 443
Destination Port: 62357
Length: 48
Checksum: 0x0a8c [unverified]
[Checksum Status: Unverified]
[Stream index: 20]
[Stream Packet Number: 3]
[Timestamps]
UDP payload (40 bytes)
QUIC IETF
0000 64 4a 7d 9c ba 6c 00 42 5a 4e e7 77 08 00 45 00
0010 00 44 00 00 40 00 3a 11 78 3a 23 ba e0 16 ac 18
0020 18 86 01 bb f3 95 00 30 0a 8c cd 00 00 00 01 00
0030 08 ee 8b 48 81 12 f0 3a db 00 40 16 ae 59 95 b5
0040 18 74 0c 74 80 32 b7 4b 1d bc 77 f9 3b 31 34 65
0050 a2 e4
Packet (82 bytes)
Decrypted QUIC (5 bytes)
User Datagram Protocol (udp), 8 bytes
Packets: 16612 · Displayed: 11181 (67.3%)
Profile: Default


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1709, y: 545
PC-PT
PC0
PC-PT
PC1
7960
IP Phone0
7960
IP Phone1
2960-24TT
Switch0
2811
Router1
Time: 00:21:51
Search for device
Realtime Simulation
Color Time(sec) Periodic Num Edit Delete
IP Phone1
Physical Config GUI Attributes
00:11a 03/01/9: 1002
From: 1001
IP Phone0
Physical Config GUI Attributes
00:11a 03/01/9: 1001
To: 1002
Ring Out


Server-PT
PC-PT
PC0
Printer-PT
Printer0
PC-PT
PC5
PC-PT
PC9
PC-PT
PC13
PC-PT
PC17
PC-PT
PC2
PC-PT
PC6
PC-PT
PC10
PC-PT
PC11
PC-PT
PC15
PC-PT
PC3
PC-PT
PC7
PC-PT
PC4
PC-PT
PC8
PC-PT
PC-PT
PC-PT


PC-PT
PC0
PC-PT
PC1
2960-24TT
Switch0
PC-PT
PC3
PC-PT
PC2

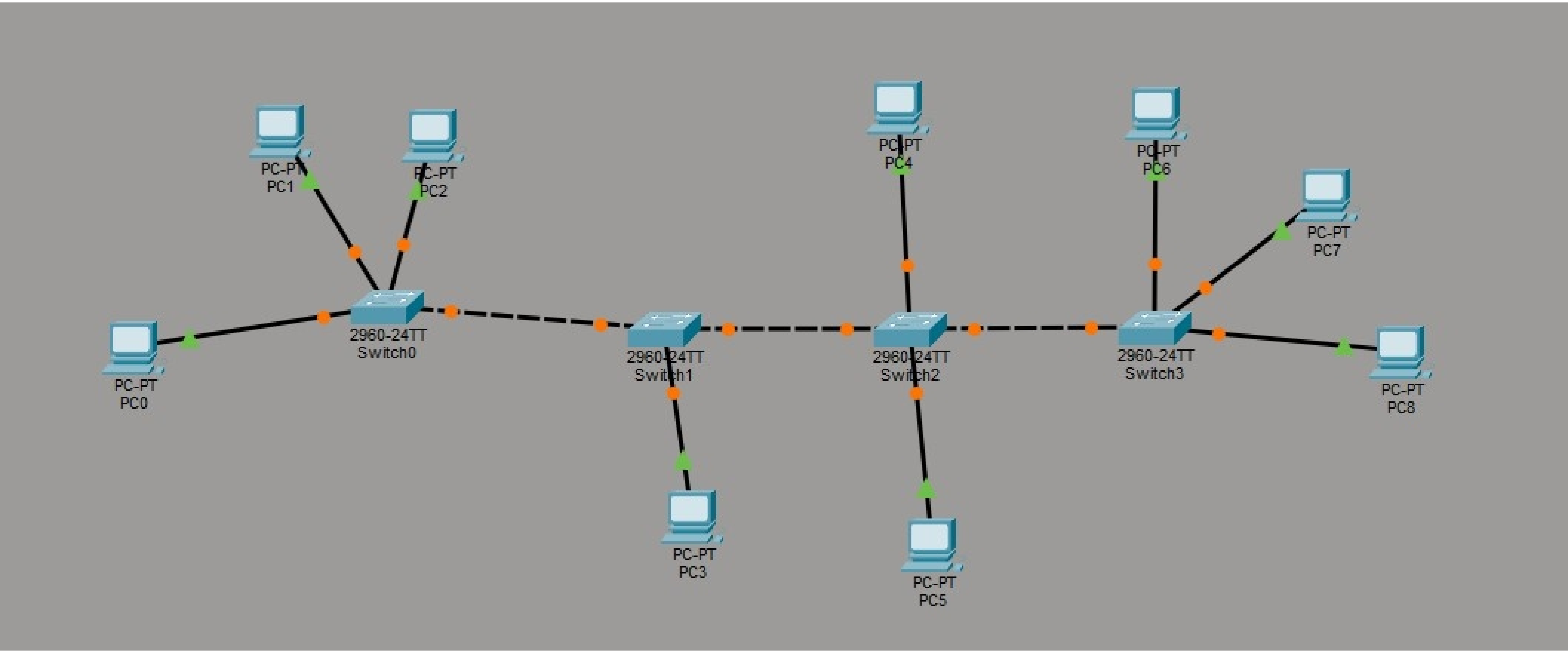
Run Debug Stop Share Save {} Beautify

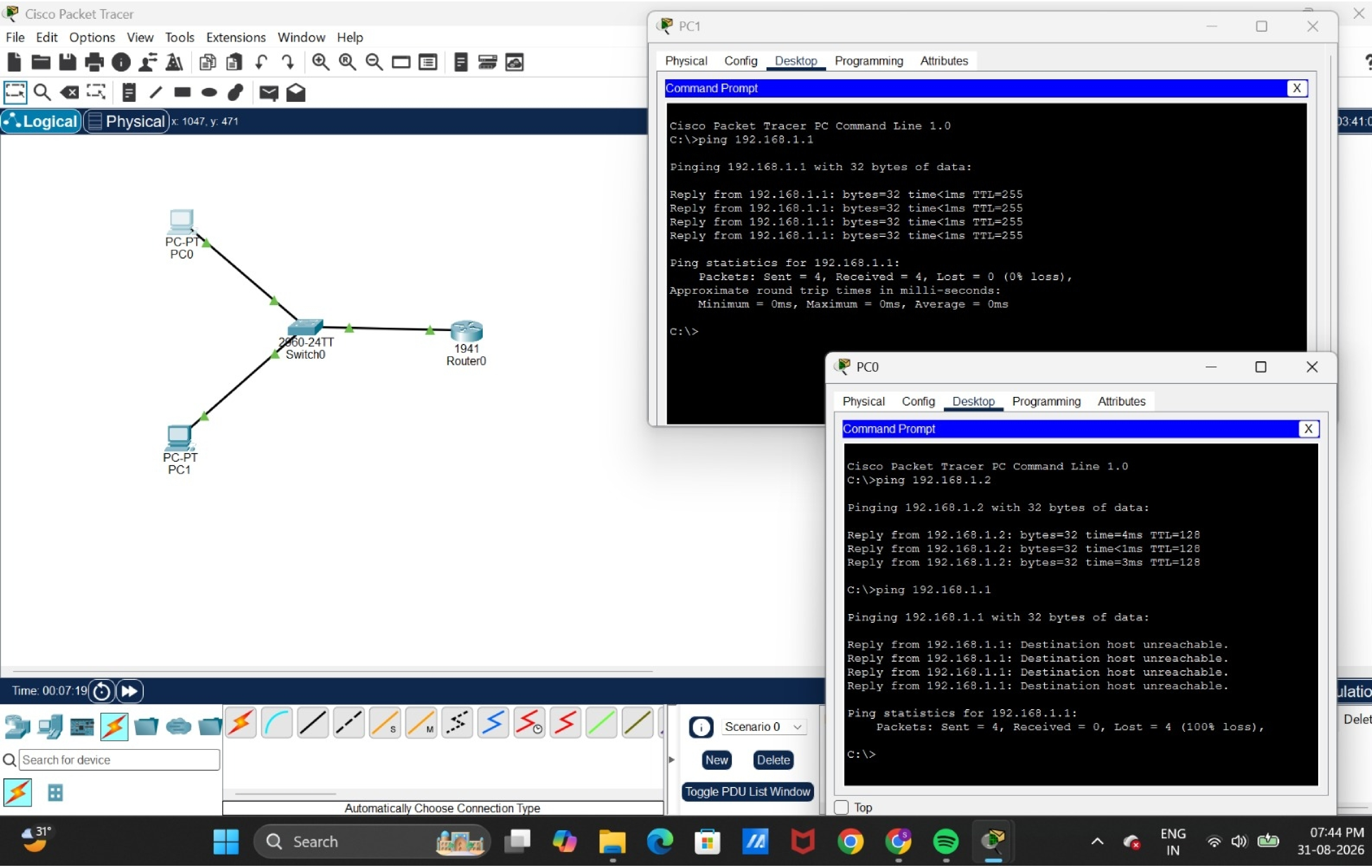
main.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 8080
#define BUFFER_SIZE 1024

int main() {
    int sock;
    struct sockaddr_in serverAddr;

    char buffer[BUFFER_SIZE];
    int bytesRead;

    FILE *file;

    sock = socket(AF_INET, SOCK_STREAM, 0);

    if (sock < 0) {
```

input

```
Connection failed


...Program finished with exit code 1
Press ENTER to exit console.
```


PC-PT
PC1
PC-PT
PC2
PC-PT
PC4
PC-PT
PC6
PC-PT
PC7
PC-PT
PC0
2960-24TT
Switch0
2960-24TT
Switch1
2960-24TT
Switch2
2960-24TT
Switch3
PC-PT
PC8
PC-PT
PC3
PC-PT
PC5


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1047, y: 471
PC-PT
PC0
2960-24TT
Switch0
1941
Router0
PC-PT
PC1
Time: 00:07:19
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top
Scenario 0
New
Delete
Toggle PDU List Window
Search for device
Automatically Choose Connection Type
Search
ENG
IN
07:44 PM
31-08-2026